

Azure management groups documentation

Azure management groups help you organize your resources and subscriptions.

About Azure Governance

OVERVIEW

[Overview of management services in Azure](#)

About Azure management groups

OVERVIEW

[Azure management groups](#)

Get started

QUICKSTART

[Create a management group \(Portal\)](#)

[Create a management group \(Azure CLI\)](#)

[Create a management group \(Azure PowerShell\)](#)

[Create a management group \(.NET\)](#)

[Create a management group \(Go\)](#)

[Create a management group \(JavaScript\)](#)

[Create a management group \(Python\)](#)

[Create a management group \(REST\)](#)

[Deploy resources to management group \(ARM templates\)](#)

HOW-TO GUIDE

[Manage your resource hierarchy](#)

[Protect your resource hierarchy](#)

Reference

REFERENCE

[Azure CLI](#)

[Azure PowerShell](#)

[Azure SDK for .NET](#)

[Azure SDK for Go](#) 

[Azure SDK for JavaScript](#)

[Azure SDK for Python](#)

[REST](#)

[Resource Manager templates](#)

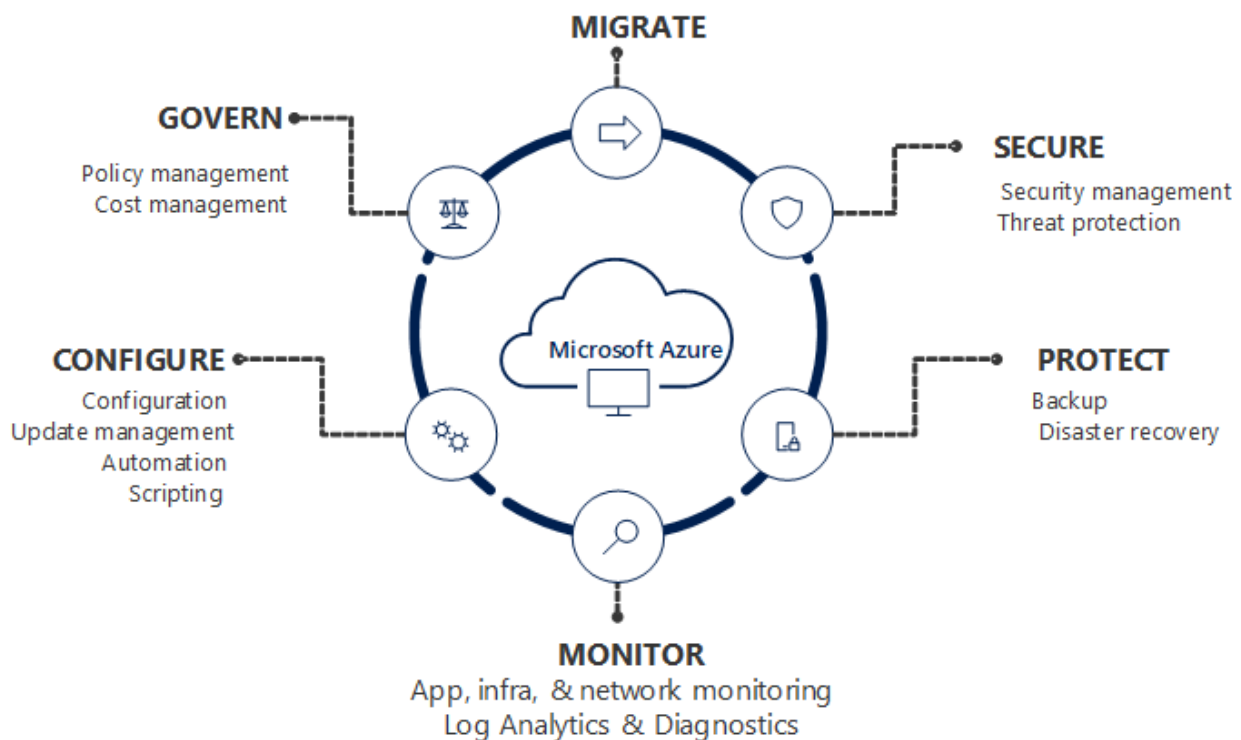
What are the Azure Management areas?

Article • 01/10/2024

Governance in Azure is one aspect of Azure Management. This article covers the different areas of management for deploying and maintaining your resources in Azure.

Management refers to the tasks and processes required to maintain your business applications and the resources that support them. Azure has many services and tools that work together to provide complete management. These services aren't only for resources in Azure, but also in other clouds and on-premises. Understanding the different tools and how they work together is the first step in designing a complete management environment.

The following diagram illustrates the different areas of management that are required to maintain any application or resource. These different areas can be thought of as a lifecycle. Each area is required in continuous succession over the lifespan of a resource. This resource lifecycle starts with the initial deployment, through continued operation, and finally when retired.



No single Azure service completely fills the requirements of a particular management area. Instead, each is realized by several services working together. Some services, such as Application Insights, provide targeted monitoring functionality for web applications. Others, like Azure Monitor logs, store management data for other services. This feature allows you to analyze data of different types collected by different services.

The following sections briefly describe the different management areas and provide links to detailed content on the main Azure services intended to address them.

Monitor

Monitoring is the act of collecting and analyzing data to audit the performance, health, and availability of your resources. An effective monitoring strategy helps you understand the operation of components and to increase your uptime with notifications. Read an overview of Monitoring that covers the different services used at [Monitoring Azure applications and resources](#).

Configure

Configure refers to the initial deployment and configuration of resources and ongoing maintenance. Automation of these tasks allows you to eliminate redundancy, minimizing your time and effort and increasing your accuracy and efficiency. [Azure Automation](#) provides the bulk of services for automating configuration tasks. While runbooks handle process automation, configuration and update management help manage configuration.

Govern

Governance provides mechanisms and processes to maintain control over your applications and resources in Azure. It involves planning your initiatives and setting strategic priorities. Governance in Azure is primarily implemented with two services. [Azure Policy](#) allows you to create, assign, and manage policy definitions to enforce rules for your resources. This feature keeps those resources in compliance with your corporate standards. [Azure Cost Management](#) allows you to track cloud usage and expenditures for your Azure resources and other cloud providers.

Secure

Manage the security of your resources and data. A security program involves assessing threats, collecting and analyzing data, and compliance of your applications and resources. Security monitoring and threat analysis are provided by [Microsoft Defender for Cloud](#), which includes unified security management and advanced threat protection across hybrid cloud workloads. See [Introduction to Azure Security](#) for comprehensive information and guidance on securing Azure resources.

Protect

Protection refers to keeping your applications and data available, even with outages that are beyond your control. Protection in Azure is provided by two services. [Azure Backup](#) provides backup and recovery of your data, either in the cloud or on-premises. [Azure Site Recovery](#) provides business continuity and immediate recovery during a disaster.

Migrate

Migration refers to transitioning workloads currently running on-premises to the Azure cloud. [Azure Migrate](#) is a service that helps you assess the migration suitability of on-premises virtual machines to Azure. Azure Site Recovery migrates virtual machines [from on-premises](#) or [from Amazon Web Services](#). [Azure Database Migration Service](#) assists you in migrating database sources to Azure Data platforms.

Next Steps

To learn more about Azure Governance, go to the following articles:

- [Azure Governance hub](#)
- [Governance in the Cloud Adoption Framework for Azure](#)

What are Azure management groups?

07/21/2025

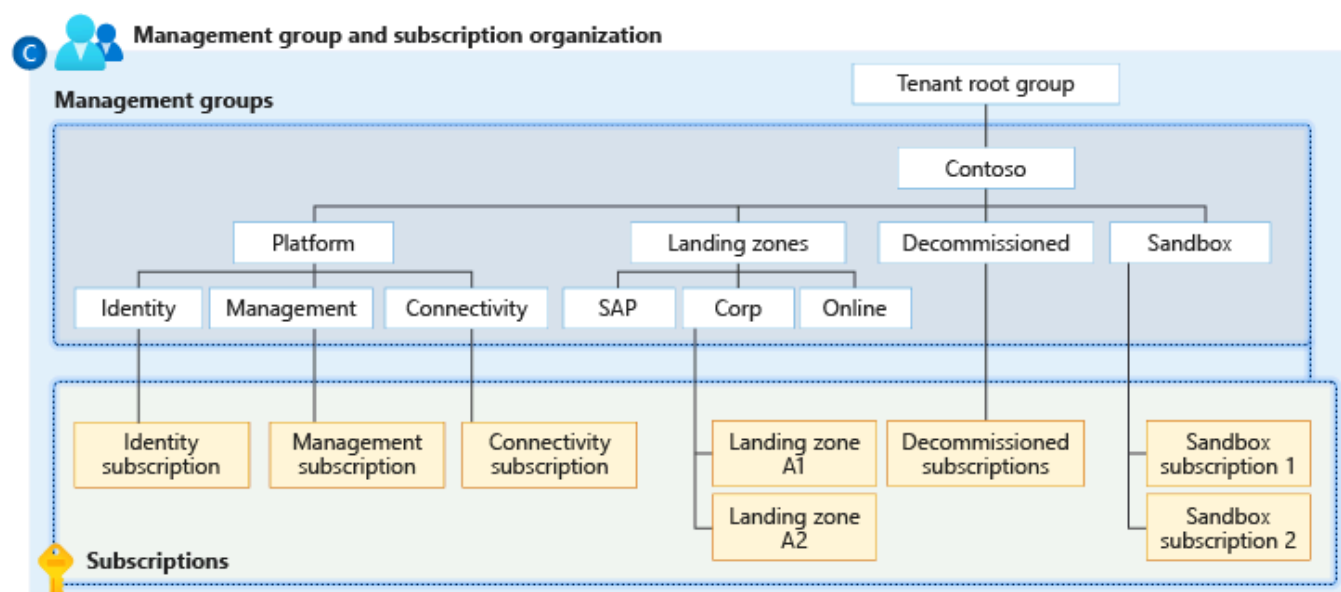
If your organization has many Azure subscriptions, you might need a way to efficiently manage access, policies, and compliance for those subscriptions. *Management groups* provide a governance scope above subscriptions. When you organize subscriptions into management groups, the governance conditions that you apply cascade by inheritance to all associated subscriptions.

Management groups give you enterprise-grade management at scale, no matter what type of subscriptions you might have. However, all subscriptions within a single management group must trust the same Microsoft Entra tenant.

For example, you can apply a policy to a management group that limits the regions available for virtual machine (VM) creation. This policy would be applied to all nested management groups, subscriptions, and resources to allow VM creation only in authorized regions.

Hierarchy of management groups and subscriptions

You can build a flexible structure of management groups and subscriptions to organize your resources into a hierarchy for unified policy and access management. The following diagram shows an example of creating a hierarchy for governance by using management groups.



You can create a hierarchy that applies a policy, for example, that limits VM locations to the West US region in the management group called *Corp*. This policy inherits all the Enterprise Agreement (EA) subscriptions that are descendants of that management group and applies to

all VMs under those subscriptions. The resource or subscription owner can't alter this security policy, to allow for improved governance.

 Note

Management groups aren't currently supported in cost management features for Microsoft Customer Agreement (MCA) subscriptions.

Another scenario where you would use management groups is to provide user access to multiple subscriptions. By moving multiple subscriptions under a management group, you can create one [Azure role assignment](#) on the management group. The role inherits that access to all the subscriptions. One assignment on the management group can enable users to have access to everything they need, instead of scripting Azure role-based access control (RBAC) over different subscriptions.

Scenario comparison

 Expand table

Scenario	Resource Group	Subscription	Management Group	Service Group	Tags
Require Inheritance from assignment on scope to each member/descendant resource	Supported*	Supported	Supported	Not Supported	Not Supported
Consolidation of resources for reduction of Role Assignments/Policy Assignments	Supported	Supported	Supported	Not Supported	Not Supported
Grouping of resources that are shared across scope boundaries. Ex. Global Networking resources in one subscription/resource group that are shared across multiple applications that have their own subscriptions/resource groups.	Not Supported	Not Supported	Not Supported	Supported	Supported
Create separate groupings that allow for separate	Not Supported	Supported	Supported	Supported	Supported**

Scenario	Resource Group	Subscription	Management Group	Service Group	Tags
aggregations of metrics					
Enforce enterprise-wide restrictions or organizational configurations across many resources	Supported*	Supported*	Supported*	Not Supported	Supported***

*: When a policy is applied to a scope, the enforcement is to all of the members within the scope. For example, on a Resource Group it only applies to the resources under it.

**: Tags can be applied across scopes and are added to resources individually. Azure Policy has built-in policies that can help manage tags.

***: Azure tags can be used as criteria within Azure Policy to apply policies to certain resources. Azure tags are subject to limitations.

Important facts about management groups

- A single directory can support 10,000 management groups.
- A management group tree can support up to six levels of depth.

This limit doesn't include the root level or the subscription level.

- Each management group and subscription can support only one parent.
- Each management group can have many children.
- All subscriptions and management groups are within a single hierarchy in each directory. For more information, see [Important facts about the root management group](#) later in this article.

Root management group for each directory

Each directory has a single top-level management group called the *root* management group. The root management group is built into the hierarchy to have all management groups and subscriptions fold up to it.

The root management group allows for the application of global policies and Azure role assignments at the directory level. Initially, the [Elevate access to manage all Azure subscriptions and management groups](#) to the User Access Administrator role of this root group. After the tenant administrator elevates access, the administrator can assign any Azure role to other

directory users or groups to manage the hierarchy. As an administrator, you can assign your account as the owner of the root management group.

Important facts about the root management group

- By default, the root management group's display name is **Tenant root group**, and it operates itself as a management group. The ID is the same value as the Microsoft Entra tenant ID.
- To change the display name, your account must have the Owner or Contributor role on the root management group. For more information, see [Change the name of a management group](#).
- The root management group can't be moved or deleted, unlike other management groups.
- All subscriptions and management groups fold up into one root management group within the directory.
 - All resources in the directory fold up to the root management group for global management.
 - New subscriptions automatically default to the root management group when they're created.
- All Azure customers can see the root management group, but not all customers have access to manage that root management group.
 - Everyone who has access to a subscription can see the context of where that subscription is in the hierarchy.
 - No one has default access to the root management group. Microsoft Entra Global Administrators are the only users who can elevate themselves to gain access. After they have access to the root management group, they can assign any Azure role to other users to manage the group.

Important

Any assignment of user access or policy on the root management group applies to all resources within the directory. Because of this access level, all customers should evaluate the need to have items defined on this scope. User access and policy assignments should be "must have" only at this scope.

Initial setup of management groups

When any user starts using management groups, an initial setup process happens. The first step is creation of the root management group in the directory. All existing subscriptions that exist in the directory then become children of the root management group.

The reason for this process is to make sure there's only one management group hierarchy within a directory. The single hierarchy within the directory allows administrative customers to apply global access and policies that other customers within the directory can't bypass.

Anything assigned on the root applies to the entire hierarchy. That is, it applies to all management groups, subscriptions, resource groups, and resources within that Microsoft Entra tenant.

Management group access

Azure management groups support [Azure RBAC](#) for all resource access and role definitions. Child resources that exist in the hierarchy inherit these permissions. Any Azure role can be assigned to a management group that inherits down the hierarchy to the resources.

For example, you can assign the Azure role VM Contributor to a management group. This role has no action on the management group but inherits to all VMs under that management group.

The following chart shows the list of roles and the supported actions on management groups.

[Expand table](#)

Azure role name	Create	Rename	Move**	Delete	Assign access	Assign policy	Read
Owner	X	X	X	X	X	X	X
Contributor	X	X	X	X			X
Management Group Contributor*	X	X	Moving Details	X			X
Reader							X
Management Group Reader*							X
Resource Policy Contributor						X	
User Access Administrator					X	X	

*: These roles allow users to perform the specified actions only on the management group scope.

****:** Role assignments on the root management group aren't required to move a subscription or a management group to and from it.

Moving subscriptions and management groups

Moving subscriptions and management groups requires different role assignments to be applied. To move a child subscription or management group the following permissions are needed:

- The child subscription or management group being moved
 - `Microsoft.management/managementgroups/write`
 - `Microsoft.management/managementgroups/subscriptions/write` (only for subscriptions)
 - `Microsoft.Authorization/roleAssignments/write`
 - `Microsoft.Authorization/roleAssignments/delete`
 - `Microsoft.Management/register/action`
- Target parent management group
 - `Microsoft.management/managementgroups/write`
- Current parent management group
 - `Microsoft.management/managementgroups/write`

For more information on moving items within the hierarchy, see [Manage your resources with management groups](#).

Azure custom role definition and assignment

You can define a management group as an assignable scope in an Azure custom role definition. The Azure custom role is available for assignment on that management group and any management group, subscription, resource group, or resource under it. The custom role inherits down the hierarchy like any built-in role.

For information about the limitations with custom roles and management groups, see [Limitations](#) later in this article.

Example definition

[Defining and creating a custom role](#) doesn't change with the inclusion of management groups. Use the full path to define the management group:

```
/providers/Microsoft.Management/managementgroups/{_groupId_}.
```

Use the management group's ID and not the management group's display name. This common error happens because both are custom-defined fields in creating a management group.

```

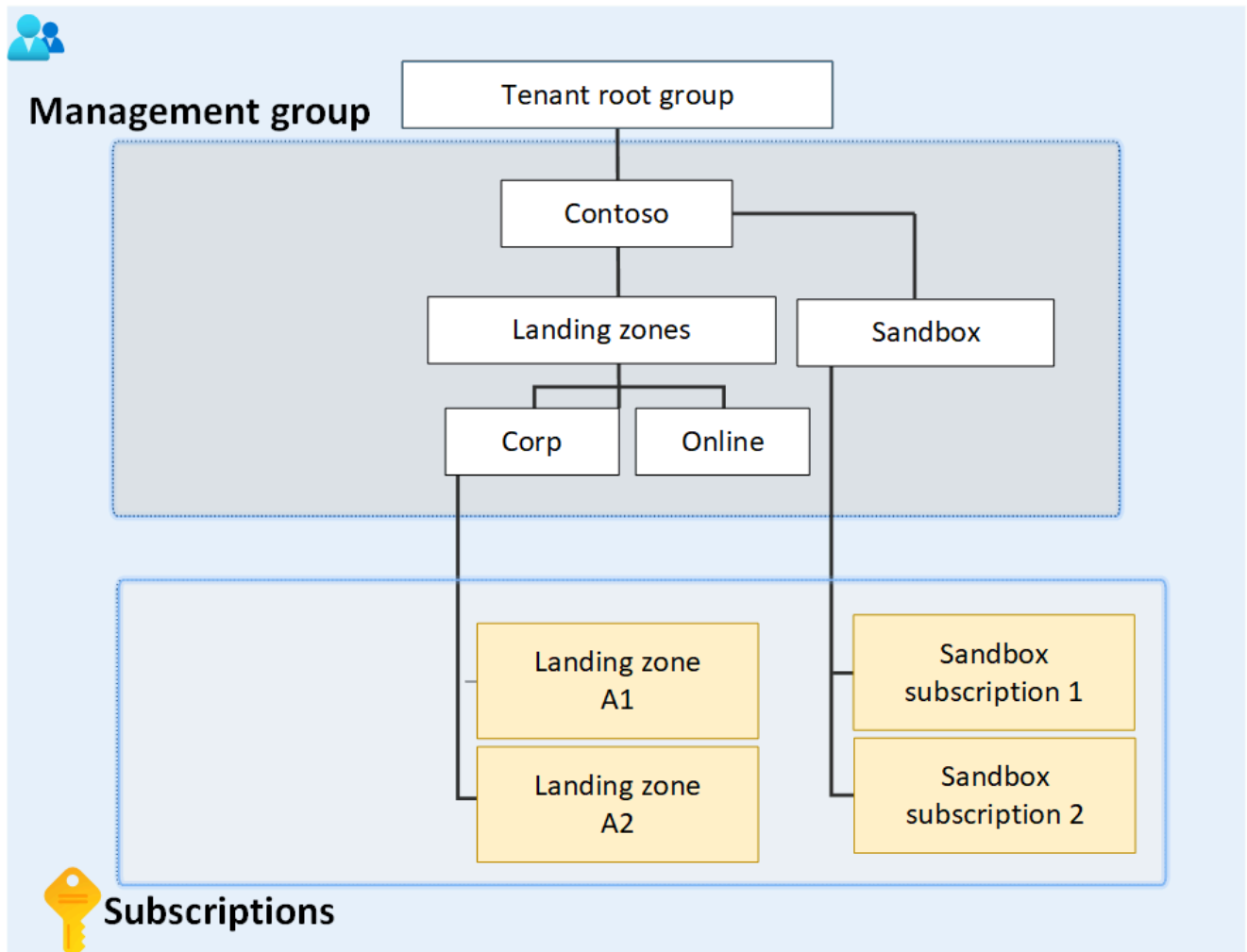
...
{
  "Name": "MG Test Custom Role",
  "Id": "id",
  "IsCustom": true,
  "Description": "This role provides members understand custom roles.",
  "Actions": [
    "Microsoft.Management/managementGroups/delete",
    "Microsoft.Management/managementGroups/read",
    "Microsoft.Management/managementGroups/write",
    "Microsoft.Management/managementGroups/subscriptions/delete",
    "Microsoft.Management/managementGroups/subscriptions/write",
    "Microsoft.resources/subscriptions/read",
    "Microsoft.Authorization/policyAssignments/*",
    "Microsoft.Authorization/policyDefinitions/*",
    "Microsoft.Authorization/policySetDefinitions/*",
    "Microsoft.PolicyInsights/*",
    "Microsoft.Authorization/roleAssignments/*",
    "Microsoft.Authorization/roledefinitions/*"
  ],
  "NotActions": [],
  "DataActions": [],
  "NotDataActions": [],
  "AssignableScopes": [
    "/providers/microsoft.management/managementGroups/ContosoCorporate"
  ]
}
...

```

Issues with breaking the role definition and assignment hierarchy path

Role definitions are assignable scopes anywhere within the management group hierarchy. A role definition can be on a parent management group, whereas the actual role assignment exists on the child subscription. Because there's a relationship between the two items, an error is seen if you try to separate the assignment from its definition.

For example, consider the following example of a small section of a hierarchy.



Assume that a custom role is defined on the sandbox management group. That custom role is then assigned on the two sandbox subscriptions.

If you try to move one of those subscriptions to be a child of the Corp management group, this breaks the path from subscription role assignment to the role definition for the sandbox management group. In this scenario, an error is received that says the move isn't allowed because it breaks this relationship.

To fix this scenario, you have these options:

- Remove the role assignment from the subscription before moving the subscription to a new parent management group.
- Add the subscription to the role definition's assignable scope.
- Change the assignable scope within the role definition. In this example, you can update the assignable scopes from the sandbox management group to the root management group so that both branches of the hierarchy can reach the definition.
- Create another custom role is defined in the other branch. This new role also requires you to change the role on the subscription.

Limitations

There are limitations to using custom roles on management groups:

- You can define only one management group in the assignable scopes of a new role. This limitation is in place to reduce the number of situations where role definitions and role assignments are disconnected. This kind of situation happens when a subscription or management group with a role assignment moves to a different parent that doesn't have the role definition.
- Custom roles with `DataActions` can't be assigned at the management group scope. For more information, see [Custom role limits](#).
- Azure Resource Manager doesn't validate the management group's existence in the role definition's assignable scope. If there's a typo or an incorrect management group ID, the role definition is still created.

Moving management groups and subscriptions

To move a management group or subscription to be a child of another management group, you need:

- Management group write permissions and role assignment write permissions on the child subscription or management group.
 - Built-in role example: Owner
- Management group write access on the target parent management group.
 - Built-in role example: Owner, Contributor, Management Group Contributor
- Management group write access on the existing parent management group.
 - Built-in role example: Owner, Contributor, Management Group Contributor

There's an exception: if the target or the existing parent management group is the root management group, the permission requirements don't apply. Because the root management group is the default landing spot for all new management groups and subscriptions, you don't need permissions on it to move an item.

If the Owner role on the subscription is inherited from the current management group, your move targets are limited. You can move the subscription only to another management group where you have the Owner role. You can't move the subscription to a management group where you're only a Contributor because you would lose ownership of the subscription. If you're directly assigned to the Owner role for the subscription, you can move it to any management group where you have the Contributor role.

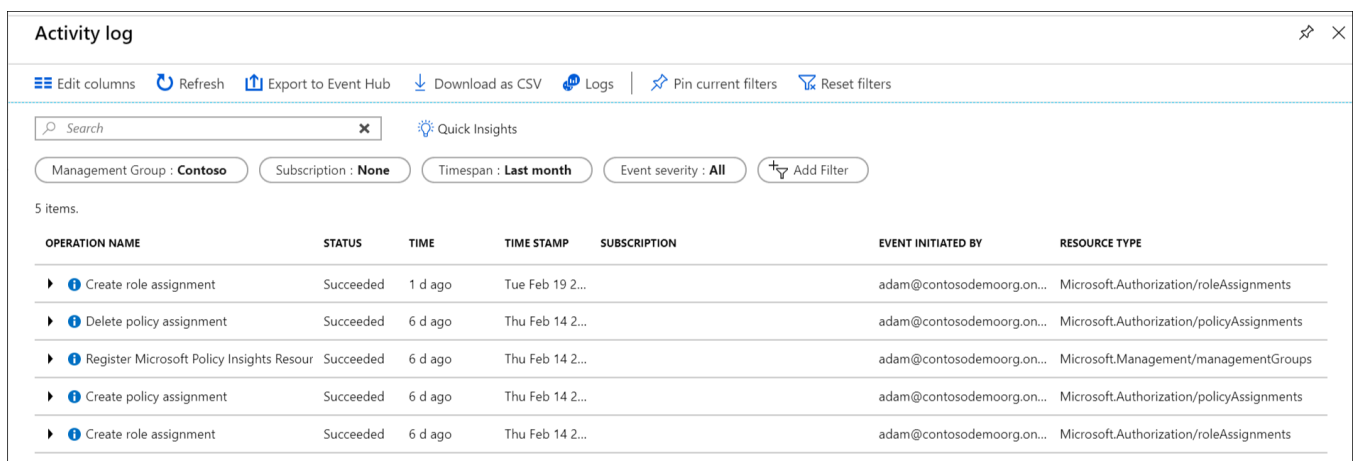
Important

Azure Resource Manager caches details of the management group hierarchy for up to 30 minutes. As a result, the Azure portal might not immediately show that you moved a

management group.

Auditing management groups by using activity logs

Management groups are supported in [Azure Monitor activity logs](#). You can query all events that happen to a management group in the same central location as other Azure resources. For example, you can see all role assignments or policy assignment changes made to a particular management group.



The screenshot shows the 'Activity log' interface in the Azure portal. At the top, there are navigation links: 'Edit columns', 'Refresh', 'Export to Event Hub', 'Download as CSV', 'Logs', 'Pin current filters', and 'Reset filters'. Below these is a search bar and a 'Quick Insights' link. The filters section shows 'Management Group : Contoso', 'Subscription : None', 'Timespan : Last month', and 'Event severity : All'. Below the filters, it says '5 items.' and displays a table of activity log entries.

OPERATION NAME	STATUS	TIME	TIME STAMP	SUBSCRIPTION	EVENT INITIATED BY	RESOURCE TYPE
▶ Create role assignment	Succeeded	1 d ago	Tue Feb 19 2...		adam@contosodemoorg.on...	Microsoft.Authorization/roleAssignments
▶ Delete policy assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/policyAssignments
▶ Register Microsoft Policy Insights Resour	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Management/managementGroups
▶ Create policy assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/policyAssignments
▶ Create role assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/roleAssignments

When you want to query on management groups outside the Azure portal, the target scope for management groups looks like

```
"/providers/Microsoft.Management/managementGroups/{management-group-id}"
```

ⓘ Note

By using the Azure Resource Manager REST API, you can enable diagnostic settings on a management group to send related Azure Monitor activity log entries to a Log Analytics workspace, Azure Storage, or Azure Event Hubs. For more information, see [Management group diagnostic settings: Create or update](#).

Related content

To learn more about management groups, see:

- [Create management groups to organize Azure resources](#)
- [Change, delete, or manage your management groups](#)
- [Protect your resource hierarchy](#)

Quickstart: Create a management group

Article • 09/03/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an Owner role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. When the Root Management Group when is created, users don't have access to it. To start using management groups, the service allows the creation of the initial management groups at the root level. For more information, see [Root management group for each directory](#).

Create in portal

1. Log into the [Azure portal](#) .
2. Select **All services > Management + governance**.
3. Select **Management Groups**.
4. Select **+ Add management group**.

Dashboard > Management groups

Search (Ctrl+/) Add management group Refresh Feedback

Click on a management group or subscription count to view and add a subscription

Search by name or ID

Showing 17 out of 17 groups

Management group	ID	Child Subscriptions	Total subscriptions
ManagementGroup-GUID	(ManagementGroup-GUID)	0	18
Contoso Marketing	ContosoMarketing	0	10
Contoso Storefront	ContosoStorefront	0	1
Contoso IT	ContosoIT	0	7

5. Leave **Create new** selected and fill in the management group ID field.

- The **Management Group ID** is the directory unique identifier that is used to submit commands on this management group. This identifier isn't editable after creation as it's used throughout the Azure system to identify this group. The **root management group** is automatically created with an ID that is the Microsoft Entra ID. For all other management groups, assign a unique ID.
- The display name field is the name that is displayed within the Azure portal. A separate display name is an optional field when creating the management group and can be changed at any time.

Add management group

Add management group

☒ Create new
 ☐ Use existing

Management group ID (Cannot be updated after creation) *

Management group display name

6. Select **Save**.

Clean up resources

To remove the management group created, follow these steps:

1. Select **All services > Management + governance**.
2. Select **Management Groups**.

3. Find the management group created above, select it, then select **Details** next to the name. Then select **Delete** and confirm the prompt.

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Quickstart: Create a management group with the Azure CLI

Article • 09/03/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- This quickstart requires that you run Azure CLI version 2.0.76 or later to install and use the CLI locally. To find the version, run `az --version`. If you need to install or upgrade, see [Install Azure CLI](#).
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an Owner role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. When the Root Management Group when is created, users don't have access to it. To start using management groups, the service allows the creation of the initial management groups at the root level. For more information, see [Root management group for each directory](#).

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

 Expand table

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com  , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal  .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Create in the Azure CLI

For Azure CLI, use the `az account management-group create` command to create a new management group. In this example, the management group **name** is *Contoso*.

Azure CLI

```
az account management-group create --name 'Contoso'
```

The **name** is a unique identifier being created. This ID is used by other commands to reference this group and it can't be changed later.

If you want the management group to show a different name within the Azure portal, add the **display-name** parameter. For example, to create a management group with the GroupName of Contoso and the display name of "Contoso Group", use the following command:

Azure CLI

```
az account management-group create --name 'Contoso' --display-name 'Contoso Group'
```

In the preceding examples, the new management group is created under the root management group. To specify a different management group as the parent, use the **parent** parameter and provide the name of the parent group.

Azure CLI

```
az account management-group create --name 'ContosoSubGroup' --parent 'Contoso'
```

Clean up resources

To remove the management group created above, use the [az account management-group delete](#) command:

Azure CLI

```
az account management-group delete --name 'Contoso'
```

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?

☐ Yes

☐ No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Quickstart: Create a management group with Azure PowerShell

Article • 05/11/2023

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- Before you start, make sure that the latest version of Azure PowerShell is installed. See [Install Azure PowerShell module](#) for detailed information.
- Any Azure AD user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an "Owner" role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. No users have access to the Root Management Group when it's created. To avoid the hurdle of finding the Azure AD Global Admins to start using management groups, we allow the creation of the initial management groups at the root level.

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	 
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Create in Azure PowerShell

For PowerShell, use the [New-AzManagementGroup](#) cmdlet to create a new management group. In this example, the management group **GroupName** is *Contoso*.

Azure PowerShell

```
New-AzManagementGroup -GroupName 'Contoso'
```

The **GroupName** is a unique identifier being created. This ID is used by other commands to reference this group and it can't be changed later.

If you want the management group to show a different name within the Azure portal, add the **DisplayName** parameter. For example, to create a management group with the GroupName of Contoso and the display name of "Contoso Group", use the following cmdlet:

Azure PowerShell

```
New-AzManagementGroup -GroupName 'Contoso' -DisplayName 'Contoso Group'
```

In the preceding examples, the new management group is created under the root management group. To specify a different management group as the parent, use the **ParentId** parameter.

Azure PowerShell

```
$parentGroup = Get-AzManagementGroup -GroupName Contoso  
New-AzManagementGroup -GroupName 'ContosoSubGroup' -ParentId $parentGroup.id
```

Clean up resources

To remove the management group created above, use the [Remove-AzManagementGroup](#) cmdlet:

Azure PowerShell

```
Remove-AzManagementGroup -GroupName 'Contoso'
```

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Quickstart: Create a management group with .NET Core

Article • 08/17/2021

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

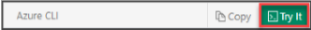
- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- An Azure service principal, including the *clientId* and *clientSecret*. If you don't have a service principal for use with Azure Policy or want to create a new one, see [Azure management libraries for .NET authentication](#). Skip the step to install the .NET Core packages as we'll do that in the next steps.
- Any Azure AD user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an "Owner" role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. No users have access to the Root Management Group when it's created. To avoid the hurdle of finding the Azure AD Global Admins to start using management groups, we allow the creation of the initial management groups at the root level.

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work

with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Application setup

To enable .NET Core to manage management groups, create a new console application and install the required packages.

1. Check that the latest .NET Core is installed (at least **3.1.8**). If it isn't yet installed, download it at dotnet.microsoft.com.
2. Initialize a new .NET Core console application named "mgCreate":

```
.NET CLI
```

```
dotnet new console --name "mgCreate"
```

3. Change directories into the new project folder and install the required packages for Azure Policy:

.NET CLI

```
# Add the Azure Policy package for .NET Core
dotnet add package Microsoft.Azure.Management.ManagementGroups --
version 1.1.1-preview

# Add the Azure app auth package for .NET Core
dotnet add package Microsoft.Azure.Services.AppAuthentication --version
1.6.1
```

4. Replace the default `program.cs` with the following code and save the updated file:

C#

```
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Microsoft.IdentityModel.Clients.ActiveDirectory;
using Microsoft.Rest;
using Microsoft.Azure.Management.ManagementGroups;
using Microsoft.Azure.Management.ManagementGroups.Models;

namespace mgCreate
{
    class Program
    {
        static async Task Main(string[] args)
        {
            string strTenant = args[0];
            string strClientId = args[1];
            string strClientSecret = args[2];
            string strGroupId = args[3];
            string strDisplayName = args[4];

            var authContext = new
AuthenticationContext($"https://login.microsoftonline.com/{strTenant}")
;

            var authResult = await authContext.AcquireTokenAsync(
                "https://management.core.windows.net",
                new ClientCredential(strClientId, strClientSecret));

            using (var client = new ManagementGroupsAPIClient(new
TokenCredentials(authResult.AccessToken)))
            {
                var mgRequest = new CreateManagementGroupRequest
                {
                    DisplayName = strDisplayName
                };
                var response = await
client.ManagementGroups.CreateOrUpdateAsync(strGroupId, mgRequest);
            }
        }
    }
}
```

```
}  
}
```

5. Build and publish the `mgCreate` console application:

.NET CLI

```
dotnet build  
dotnet publish -o {run-folder}
```

Create the management group

In this quickstart, you create a new management group in the root management group.

1. Change directories to the `{run-folder}` you defined with the previous `dotnet publish` command.
2. Enter the following command in the terminal:

Bash

```
mgCreate.exe `  
    "{tenantId}" `  
    "{clientId}" `  
    "{clientSecret}" `  
    "{groupId}" `  
    "{displayName}"`
```

The preceding commands use the following information:

- `{tenantId}` - Replace with your tenant ID
- `{clientId}` - Replace with the client ID of your service principal
- `{clientSecret}` - Replace with the client secret of your service principal
- `{groupId}` - Replace with the ID for your new management group
- `{displayName}` - Replace with the friendly name for your new management group

The result is a new management group in the root management group.

Clean up resources

- Delete the new management group through the portal.

- If you wish to remove the .NET Core console application and installed packages, delete the `mgCreate` project folder.

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Quickstart: Create a management group with Go

Article • 09/03/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- An Azure service principal, including the *clientId* and *clientSecret*. If you don't have a service principal for use with Azure Policy or want to create a new one, see [Azure management libraries for .NET authentication](#). Skip the step to install the .NET Core packages as we'll do that in the next steps.
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an Owner role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. When the Root Management Group when is created, users don't have access to it. To start using management groups, the service allows the creation of the initial management groups at the root level. For more information, see [Root management group for each directory](#).

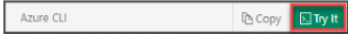
Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work

with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

 Expand table

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Add the management group package

To enable Go to manage management groups, the package must be added. This package works wherever Go can be used, including [bash on Windows 10](#) or locally installed.

1. Check that the latest Go is installed (at least **1.15**). If it isn't yet installed, download it at [Golang.org](https://golang.org).
2. Check that the latest Azure CLI is installed (at least **2.5.1**). If it isn't yet installed, see [Install the Azure CLI](#).

 **Note**

Azure CLI is required to enable Go to use the `auth.NewAuthorizerFromCLI()` method in the following example. For information about other options, see [Azure SDK for Go - More authentication details](#) [↗].

3. Authenticate through Azure CLI.

Azure CLI

```
az login
```

4. In your Go environment of choice, install the required packages for management groups:

Bash

```
# Add the management group package for Go
go install github.com/Azure/azure-sdk-for-go/services/resources/mgmt/2020-05-01/managementgroups@latest

# Add the Azure auth package for Go
go install github.com/Azure/go-autorest/autorest/azure/auth@latest
```

Application setup

With the Go packages added to your environment of choice, it's time to set up the Go application that can create a management group.

1. Create the Go application and save the following source as `mgCreate.go`:

Go

```
package main

import (
    "context"
    "fmt"
    "os"

    mg "github.com/Azure/azure-sdk-for-go/services/resources/mgmt/2020-05-01/managementgroups"
    "github.com/Azure/go-autorest/autorest/azure/auth"
)

func main() {
    // Get variables from command line arguments
    var mgName = os.Args[1]
```



```

// Create and authorize a client
mgClient := mg.NewClient()
authorizer, err := auth.NewAuthorizerFromCLI()
if err == nil {
    mgClient.Authorizer = authorizer
} else {
    fmt.Printf(err.Error())
}

// Create the request
Request := mg.CreateManagementGroupRequest{
    Name: &mgName,
}

// Run the query and get the results
var results, queryErr = mgClient.CreateOrUpdate(context.Background(),
mgName, Request, "no-cache")
if queryErr == nil {
    fmt.Printf("Results: " + fmt.Sprint(results) + "\n")
} else {
    fmt.Printf(queryErr.Error())
}
}

```

2. Build the Go application:

Bash

```
go build mgCreate.go
```

3. Create a management group using the compiled Go application. Replace `<Name>` with the name of your new management group:

Bash

```
mgCreate "<Name>"
```

The result is a new management group in the root management group.

Clean up resources

If you wish to remove the installed packages from your Go environment, you can do so by using the following command:

Bash

```
# Remove the installed packages from the Go environment
go clean -i github.com/Azure/azure-sdk-for-go/services/resources/mgmt/2020-05-01/managementgroups
go clean -i github.com/Azure/go-autorest/autorest/azure/auth
```

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Quickstart: Create a management group with JavaScript

Article • 09/03/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- Before you start, make sure that at least version 12 of [Node.js](#)  is installed.
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an Owner role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. When the Root Management Group when is created, users don't have access to it. To start using management groups, the service allows the creation of the initial management groups at the root level. For more information, see [Root management group for each directory](#).

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Application setup

To enable JavaScript to manage management groups, the environment must be set up. This setup works wherever JavaScript can be used, including [bash on Windows 10](#).

1. Set up a new Node.js project by running the following command.

Bash

```
npm init -y
```

2. Add a reference to the yargs module.

Bash

```
npm install yargs
```

3. Add a reference to the Azure Resource Graph module.

Bash

```
npm install @azure/arm-managementgroups
```

4. Add a reference to the Azure authentication library.

Bash

```
npm install @azure/identity
```

ⓘ Note

Verify in *package.json* `@azure/arm-managementgroups` is version **2.0.1** or higher and `@azure/identity` is version **2.0.4** or higher.

Create the management group

1. Create a new file named *index.js* and enter the following code.

JavaScript

```
const argv = require("yargs").argv;
const { InteractiveBrowserCredential } = require("@azure/identity");
const { ManagementGroupsAPI } = require("@azure/arm-
managementgroups");

if (argv.groupID && argv.displayName) {
  const createMG = async () => {
    const credentials = new InteractiveBrowserCredential();
    const client = new ManagementGroupsAPI(credentials);
    const result = await
client.managementGroups.beginCreateOrUpdateAndWait(
  argv.groupID,
  {
    displayName: argv.displayName
  }
);
    console.log(result);
  };

  createMG();
}
```

2. Enter the following command in the terminal:

Bash

```
node index.js --groupID "<NEW_MG_GROUP_ID>" --displayName "  
<NEW_MG_FRIENDLY_NAME>"
```

Make sure to replace each token `<>` placeholder with your *management group ID* and *management group friendly name*, respectively.

As the script attempts to authenticate, a message similar to the following message is displayed in the terminal:

To sign in, use a web browser to open the page
<https://microsoft.com/devicelogin> and enter the code FGB56WJUGK to
authenticate.

Once you authenticate in the browser, then the script continues to run.

The result of creating the management group is output to the console.

Clean up resources

If you wish to remove the installed libraries from your application, run the following command.

Bash

```
npm uninstall @azure/arm-managementgroups @azure/identity yargs
```

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?



Yes



No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Quickstart: Create a management group with Python

Article • 09/03/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an Owner role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. When the Root Management Group when is created, users don't have access to it. To start using management groups, the service allows the creation of the initial management groups at the root level. For more information, see [Root management group for each directory](#).

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Add the Resource Graph library

To enable Python to manage management groups, the library must be added. This library works wherever Python can be used, including [bash on Windows 10](#) or locally installed.

1. Check that the latest Python is installed (at least **3.8**). If it isn't yet installed, download it at [Python.org](#).
2. Check that the latest Azure CLI is installed (at least **2.5.1**). If it isn't yet installed, see [Install the Azure CLI](#).

ⓘ Note

Azure CLI is required to enable Python to use the **CLI-based authentication** in the following examples. For information about other options, see [Authenticate using the Azure management libraries for Python](#).

3. Authenticate through Azure CLI.

Azure CLI

```
az login
```

4. In your Python environment of choice, install the required libraries for management groups:

Bash

```
# Add the management groups library for Python
pip install azure-mgmt-managementgroups

# Add the Resources library for Python
pip install azure-mgmt-resource

# Add the CLI Core library for Python for authentication (development only!)
pip install azure-cli-core
```

ⓘ Note

If Python is installed for all users, these commands must be run from an elevated console.

5. Validate that the libraries have been installed. `azure-mgmt-managementgroups` should be **0.2.0** or higher, `azure-mgmt-resource` should be **9.0.0** or higher, and `azure-cli-core` should be **2.5.0** or higher.

Bash

```
# Check each installed library
pip show azure-mgmt-managementgroups azure-mgmt-resource azure-cli-core
```

Create the management group

1. Create the Python script and save the following source as `mgCreate.py`:

Python

```
# Import management group classes
from azure.mgmt.managementgroups import ManagementGroupsAPI

# Import specific methods and models from other libraries
from azure.common.credentials import get_azure_cli_credentials
from azure.common.client_factory import get_client_from_cli_profile
from azure.mgmt.resource import ResourceManagementClient,
```

```

SubscriptionClient

# Wrap all the work in a function
def createmanagementgroup( strName ):
    # Get your credentials from Azure CLI (development only!) and get
    your subscription list
    subsClient = get_client_from_cli_profile(SubscriptionClient)
    subsRaw = []
    for sub in subsClient.subscriptions.list():
        subsRaw.append(sub.as_dict())
    subsList = []
    for sub in subsRaw:
        subsList.append(sub.get('subscription_id'))

    # Create management group client and set options
    mgClient = get_client_from_cli_profile(ManagementGroupsAPI)
    mg_request = {'name': strName, 'display_name': strName}

    # Create management group
    mg =
mgClient.management_groups.create_or_update(group_id=strName,create_man
agement_group_request=mg_request)

    # Show results
    print(mg)

createmanagementgroup("MyNewMG")

```

2. Authenticate with Azure CLI with `az login`.

3. Enter the following command in the terminal:

```
Bash
```

```
py mgCreate.py
```

The result of creating the management group is output to the console as an `LROPoller` object.

Clean up resources

If you wish to remove the installed libraries from your Python environment, you can do so by using the following command:

```
Bash
```

```

# Remove the installed libraries from the Python environment
pip uninstall azure-mgmt-managementgroups azure-mgmt-resource azure-cli-core

```

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Quickstart: Create a management group with REST API

Article • 07/19/2024

Management groups are containers that help you manage access, policy, and compliance across multiple subscriptions. Create these containers to build an effective and efficient hierarchy that can be used with [Azure Policy](#) and [Azure Role Based Access Controls](#). For more information on management groups, see [Organize your resources with Azure management groups](#).

The first management group created in the directory could take up to 15 minutes to complete. There are processes that run the first time to set up the management groups service within Azure for your directory. You receive a notification when the process is complete. For more information, see [initial setup of management groups](#).

Prerequisites

- If you don't have an Azure subscription, create a [free](#)  account before you begin.
- If you haven't already, install [ARMClient](#) . It's a tool that sends HTTP requests to Azure Resource Manager-based REST APIs.
- Any Microsoft Entra ID user in the tenant can create a management group without the management group write permission assigned to that user if [hierarchy protection](#) isn't enabled. This new management group becomes a child of the Root Management Group or the [default management group](#) and the creator is given an "Owner" role assignment. Management group service allows this ability so that role assignments aren't needed at the root level. No users have access to the Root Management Group when it's created. To avoid the hurdle of finding the Microsoft Entra ID Global Admins to start using management groups, we allow the creation of the initial management groups at the root level.

Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services. You can use the Cloud Shell preinstalled commands to run the code in this article, without having to install anything on your local environment.

To start Azure Cloud Shell:

Option	Example/Link
Select Try It in the upper-right corner of a code or command block. Selecting Try It doesn't automatically copy the code or command to Cloud Shell.	
Go to https://shell.azure.com , or select the Launch Cloud Shell button to open Cloud Shell in your browser.	
Select the Cloud Shell button on the menu bar at the upper right in the Azure portal .	

To use Azure Cloud Shell:

1. Start Cloud Shell.
2. Select the **Copy** button on a code block (or command block) to copy the code or command.
3. Paste the code or command into the Cloud Shell session by selecting **Ctrl+Shift+V** on Windows and Linux, or by selecting **Cmd+Shift+V** on macOS.
4. Select **Enter** to run the code or command.

Create in REST API

For REST API, use the [Management Groups - Create or Update](#) endpoint to create a new management group. In this example, the management group **groupId** is *Contoso*.

- REST API URI

HTTP
PUT https://management.azure.com/providers/Microsoft.Management/managementGroups/Contoso?api-version=2020-05-01

- No Request Body

The **groupId** is a unique identifier being created. This ID is used by other commands to reference this group and it can't be changed later.

If you want the management group to show a different name within the Azure portal, add the **properties.displayName** property in the request body. For example, to create a

management group with the **groupId** of *Contoso* and the display name of *Contoso Group*, use the following endpoint and request body:

- REST API URI

HTTP

PUT

<https://management.azure.com/providers/Microsoft.Management/managementGroups/Contoso?api-version=2020-05-01>

- Request Body

JSON

```
{
  "properties": {
    "displayName": "Contoso Group"
  }
}
```

In the preceding examples, the new management group is created under the root management group. To specify a different management group as the parent, use the **properties.parent.id** property.

- REST API URI

HTTP

PUT

<https://management.azure.com/providers/Microsoft.Management/managementGroups/Contoso?api-version=2020-05-01>

- Request Body

JSON

```
{
  "properties": {
    "displayName": "Contoso Group",
    "parent": {
      "id":
        "/providers/Microsoft.Management/managementGroups/HoldingGroup"
    }
  }
}
```

Clean up resources

To remove the management group created above, use the [Management Groups - Delete](#) endpoint:

- REST API URI

HTTP

DELETE

<https://management.azure.com/providers/Microsoft.Management/managementGroups/Contoso?api-version=2020-05-01>

- No Request Body

Next steps

In this quickstart, you created a management group to organize your resource hierarchy. The management group can hold subscriptions or other management groups.

To learn more about management groups and how to manage your resource hierarchy, continue to:

[Manage your resources with management groups](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Azure Resource Graph sample queries for management groups

Article • 06/12/2024

This page is a collection of [Azure Resource Graph](#) sample queries for management groups.

Sample queries

Count of subscriptions per management group

Summarizes the count of subscriptions in each management group.

Kusto

```
ResourceContainers
| where type =~ 'microsoft.management/managementgroups'
| project mgname = name
| join kind=leftouter (resourcecontainers | where type =~
'microsoft.resources/subscriptions'
| extend mgParent = properties.managementGroupAncestorsChain | project id,
mgname = tostring(mgParent[0].name)) on mgname
| summarize count() by mgname
```

Azure CLI

Azure CLI

```
az graph query -q "ResourceContainers | where type =~
'microsoft.management/managementgroups' | project mgname = name | join
kind=leftouter (resourcecontainers | where type =~
'microsoft.resources/subscriptions' | extend mgParent =
properties.managementGroupAncestorsChain | project id, mgname =
tostring(mgParent[0].name)) on mgname | summarize count() by mgname"
```

List all management group ancestors for a specified management group

Provides the management group hierarchy details for the management group specified in the [query scope](#). In this example, the management group is named **Application**.

Kusto

```
ResourceContainers
| where type =~ 'microsoft.management/managementgroups'
| extend mgParent = properties.details.managementGroupAncestorsChain
| mv-expand with_itemindex=MGHierarchy mgParent
| project name, properties.displayName, mgParent, MGHierarchy, mgParent.name
```

Azure CLI

Azure CLI

```
az graph query -q "ResourceContainers | where type =~
'microsoft.management/managementgroups' | extend mgParent =
properties.details.managementGroupAncestorsChain | mv-expand
with_itemindex=MGHierarchy mgParent | project name,
properties.displayName, mgParent, MGHierarchy, mgParent.name" --
management-groups Application
```

List all management group ancestors for a specified subscription

Provides the management group hierarchy details for the subscription specified in the [query scope](#). In this example, the subscription GUID is `11111111-1111-1111-1111-111111111111`.

Kusto

```
ResourceContainers
| where type =~ 'microsoft.resources/subscriptions'
| extend mgParent = properties.managementGroupAncestorsChain
| mv-expand with_itemindex=MGHierarchy mgParent
| project subscriptionId, name, mgParent, MGHierarchy, mgParent.name
```

Azure CLI

Azure CLI

```
az graph query -q "ResourceContainers | where type =~
'microsoft.resources/subscriptions' | extend mgParent =
properties.managementGroupAncestorsChain | mv-expand
with_itemindex=MGHierarchy mgParent | project subscriptionId, name,
```

```
mgParent, MGHierarchy, mgParent.name" --subscriptions 11111111-1111-1111-1111-111111111111
```

List all subscriptions under a specified management group

Provides the name and subscription ID of all subscriptions under the management group specified in the [query scope](#). In this example, the management group is named **Application**.

Kusto

```
ResourceContainers
| where type =~ 'microsoft.resources/subscriptions'
| project subscriptionId, name
```

Azure CLI

Azure CLI

```
az graph query -q "ResourceContainers | where type =~
'microsoft.resources/subscriptions' | project subscriptionId, name" --
management-groups Application
```

Secure score per management group

Returns secure score per management group.

Kusto

```
SecurityResources
| where type == 'microsoft.security/securescores'
| project subscriptionId,
    subscriptionTotal = iff(properties.score.max == 0, 0.00,
round(tolong(properties.weight) *
todouble(properties.score.current)/tolong(properties.score.max),2)),
    weight = tolong(iff(properties.weight == 0, 1, properties.weight))
| join kind=leftouter (
    ResourceContainers
    | where type == 'microsoft.resources/subscriptions' and properties.state
    == 'Enabled'
    | project subscriptionId, mgChain=properties.managementGroupAncestorsChain
)
on subscriptionId
```

```
| mv-expand mg=mgChain
| summarize sumSubs = sum(subscriptionTotal), sumWeight = sum(weight),
resultsNum = count() by toString(mg.displayName), mgId = toString(mg.name)
| extend secureScore = iff(tolong(resultsNum) == 0, 404.00,
round(sumSubs/sumWeight*100,2))
| project mgName=mg_displayName, mgId, sumSubs, sumWeight, resultsNum,
secureScore
| order by mgName asc
```

Azure CLI

Azure CLI

```
az graph query -q "SecurityResources | where type ==
'microsoft.security/securescores' | project subscriptionId,
subscriptionTotal = iff(properties.score.max == 0, 0.00,
round(tolong(properties.weight) *
todouble(properties.score.current)/tolong(properties.score.max),2)),
weight = tolong(iff(properties.weight == 0, 1, properties.weight)) |
join kind=leftouter ( ResourceContainers | where type ==
'microsoft.resources/subscriptions' and properties.state == 'Enabled' |
project subscriptionId, mgChain=properties.managementGroupAncestorsChain
) on subscriptionId | mv-expand mg=mgChain | summarize sumSubs =
sum(subscriptionTotal), sumWeight = sum(weight), resultsNum = count() by
toString(mg.displayName), mgId = toString(mg.name) | extend secureScore
= iff(tolong(resultsNum) == 0, 404.00, round(sumSubs/sumWeight*100,2)) |
project mgName=mg_displayName, mgId, sumSubs, sumWeight, resultsNum,
secureScore | order by mgName asc"
```

Next steps

- Learn more about the [query language](#).
- Learn more about how to [explore resources](#).

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#) 

Manage your Azure subscriptions at scale with management groups

Article • 09/06/2024

If your organization has many subscriptions, you might need a way to efficiently manage access, policies, and compliance for those subscriptions. Azure management groups provide a level of scope above subscriptions. You organize subscriptions into containers called *management groups* and apply your governance conditions to the management groups. All subscriptions within a management group automatically inherit the conditions applied to the management group.

Management groups give you enterprise-grade management at a large scale no matter what type of subscription you have. To learn more about management groups, see [Organize your resources with Azure management groups](#).

ⓘ Note

This article provides steps about how to delete personal data from the device or service and can be used to support your obligations under the GDPR. For general information about GDPR, see the [GDPR section of the Microsoft Trust Center](#) and the [GDPR section of the Service Trust portal](#).

ⓘ Important

Azure Resource Manager user tokens and management group cache last for 30 minutes before they're forced to refresh. Any action like moving a management group or subscription might take up to 30 minutes to appear. To see the updates sooner, you need to update your token by refreshing the browser, signing in and out, or requesting a new token.

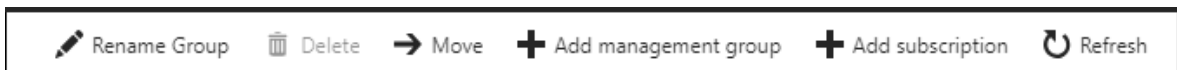
For the Azure PowerShell actions in this article, keep in mind that `AzManagementGroup`-related cmdlets mention that `-GroupId` is an alias of the `-GroupName` parameter. You can use either of them to provide the management group ID as a string value.

Change the name of a management group

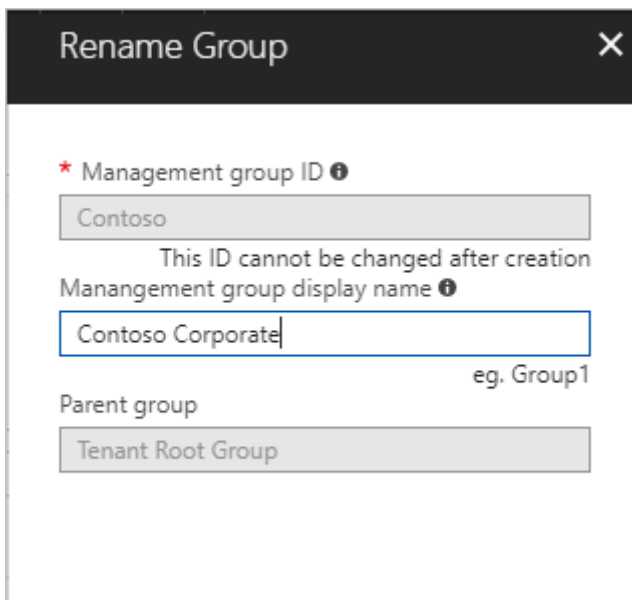
You can change the name of the management group by using the Azure portal, Azure PowerShell, or the Azure CLI.

Change the name in the portal

1. Sign in to the [Azure portal](#).
2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.
3. Select the management group that you want to rename.
4. Select **details**.
5. Select the **Rename Group** option at the top of the pane.



6. On the **Rename Group** pane, enter the new name that you want to display.

A dialog box titled 'Rename Group' with a close button (X) in the top right corner. It contains three input fields: 'Management group ID' with the value 'Contoso' and a note 'This ID cannot be changed after creation'; 'Management group display name' with the value 'Contoso Corporate' and a note 'eg. Group1'; and 'Parent group' with the value 'Tenant Root Group'.

7. Select **Save**.

Change the name in Azure PowerShell

To update the display name, use `Update-AzManagementGroup` in Azure PowerShell. For example, to change a management group's display name from **Contoso IT** to **Contoso Group**, run the following command:

Azure PowerShell

```
Update-AzManagementGroup -GroupId 'ContosoIt' -DisplayName 'Contoso Group'
```

Change the name in the Azure CLI

For the Azure CLI, use the `update` command:

Azure CLI

```
az account management-group update --name 'Contoso' --display-name 'Contoso Group'
```

Delete a management group

To delete a management group, you must meet the following requirements:

- There are no child management groups or subscriptions under the management group. To move a subscription or management group to another management group, see [Move management groups and subscriptions](#) later in this article.
- You need write permissions on the management group (Owner, Contributor, or Management Group Contributor). To see what permissions you have, select the management group and then select **IAM**. To learn more on Azure roles, see [What is Azure role-based access control \(Azure RBAC\)?](#).

Delete a management group in the portal

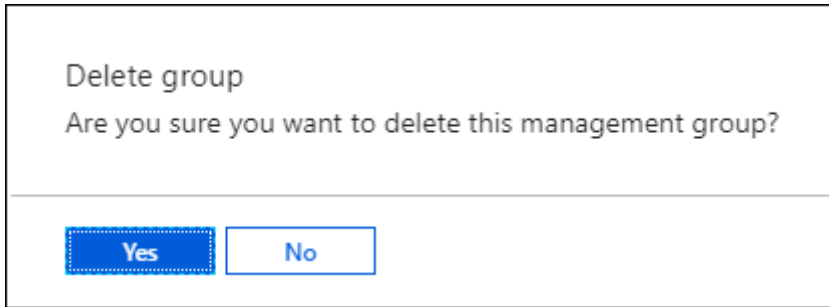
1. Sign in to the [Azure portal](#).
2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.
3. Select the management group that you want to delete.
4. Select **details**.
5. Select **Delete**.

 Rename Group	 Delete	 Move	 Add management group	 Add subscription	 Refresh
Name Newgroup1	Parent management group Contoso Redmond				
ID Newgroup	Child management groups 0				
Access Level Owner					
					
Search by name or ID					
NAME			ID		
No result					

 **Tip**

If the **Delete** button is unavailable, hovering over the button shows you the reason.

6. A dialog opens and asks you to confirm that you want to delete the management group.

A screenshot of a confirmation dialog box titled "Delete group". The text inside asks, "Are you sure you want to delete this management group?". At the bottom, there are two buttons: "Yes" (highlighted with a blue border) and "No" (with a light blue border).

7. Select **Yes**.

Delete a management group in Azure PowerShell

To delete a management group, use the `Remove-AzManagementGroup` command in Azure PowerShell:

Azure PowerShell

```
Remove-AzManagementGroup -GroupId 'Contoso'
```

Delete a management group in the Azure CLI

With the Azure CLI, use the command `az account management-group delete`:

Azure CLI

```
az account management-group delete --name 'Contoso'
```

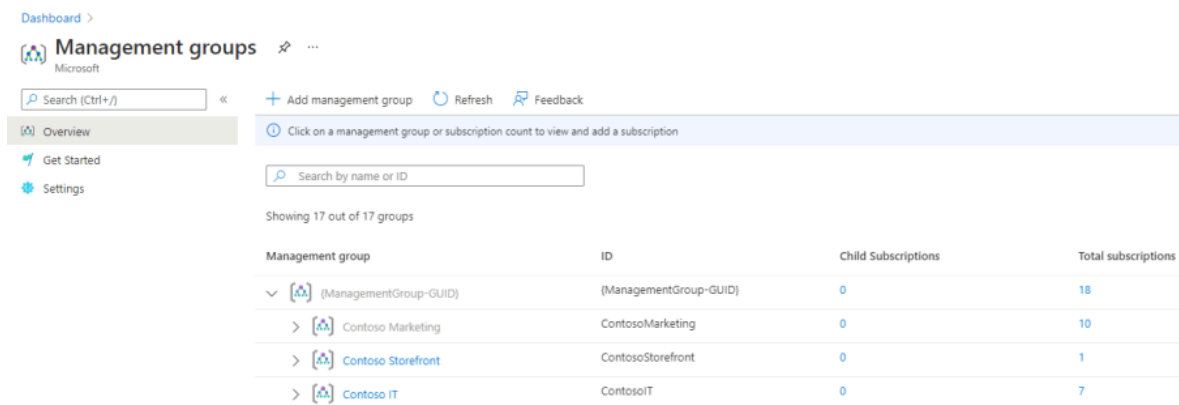
View management groups

You can view any management group if you have a direct or inherited Azure role on it.

View management groups in the portal

1. Sign in to the [Azure portal](#).

2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.
3. The page for management group hierarchy appears. On this page, you can explore all the management groups and subscriptions that you have access to. Selecting the group name takes you to a lower level in the hierarchy. The navigation works the same as a file explorer does.
4. To see the details of the management group, select the **(details)** link next to the title of the management group. If this link isn't available, you don't have permissions to view that management group.



View management groups in Azure PowerShell

You use the `Get-AzManagementGroup` command to retrieve all groups. For the full list of `GET` PowerShell commands for management groups, see the [Az.Resources](#) modules.

Azure PowerShell

```
Get-AzManagementGroup
```

For a single management group's information, use the `-GroupId` parameter:

Azure PowerShell

```
Get-AzManagementGroup -GroupId 'Contoso'
```

To return a specific management group and all the levels of the hierarchy under it, use the `-Expand` and `-Recurse` parameters:

Azure PowerShell

```
PS C:\> $response = Get-AzManagementGroup -GroupId TestGroupParent -Expand -Recurse
```

```

PS C:\> $response

Id
:
/providers/Microsoft.Management/managementGroups/TestGroupParent
Type
: /providers/Microsoft.Management/managementGroups
Name
: TestGroupParent
TenantId
: 00000000-0000-0000-0000-000000000000
DisplayName
: TestGroupParent
UpdatedTime
: 2/1/2018 11:15:46 AM
UpdatedBy
: 00000000-0000-0000-0000-000000000000
ParentId
:
/providers/Microsoft.Management/managementGroups/00000000-0000-0000-0000-
000000000000
ParentName
: 00000000-0000-0000-0000-000000000000
ParentDisplayName
: 00000000-0000-0000-0000-000000000000
Children
: {TestGroup1DisplayName, TestGroup2DisplayName}

PS C:\> $response.Children[0]

Type
: /managementGroup
Id
: /providers/Microsoft.Management/managementGroups/TestGroup1
Name
: TestGroup1
DisplayName
: TestGroup1DisplayName
Children
: {TestRecurseChild}

PS C:\> $response.Children[0].Children[0]

Type
: /managementGroup
Id
:
/providers/Microsoft.Management/managementGroups/TestRecurseChild
Name
: TestRecurseChild
DisplayName
: TestRecurseChild
Children
:

```

View management groups in the Azure CLI

You use the `list` command to retrieve all groups:

Azure CLI

```
az account management-group list
```

For a single management group's information, use the `show` command:

Azure CLI

```
az account management-group show --name 'Contoso'
```

To return a specific management group and all the levels of the hierarchy under it, use the `-Expand` and `-Recurse` parameters:

Azure CLI

```
az account management-group show --name 'Contoso' -e -r
```

Move management groups and subscriptions

One reason to create a management group is to bundle subscriptions together. Only management groups and subscriptions can become children of another management group. A subscription that moves to a management group inherits all user access and policies from the parent management group.

You can move subscriptions between management groups. A subscription can have only one parent management group.

When you move a management group or subscription to be a child of another management group, three rules need to be evaluated as true.

If you're doing the move action, you need permission at each of the following layers:

- Child subscription or management group
 - `Microsoft.management/managementgroups/write`
 - `Microsoft.management/managementgroups/subscriptions/write` (only for subscriptions)
 - `Microsoft.Authorization/roleAssignments/write`
 - `Microsoft.Authorization/roleAssignments/delete`
 - `Microsoft.Management/register/action`
- Target parent management group
 - `Microsoft.management/managementgroups/write`
- Current parent management group
 - `Microsoft.management/managementgroups/write`

There's an exception: if the target or the existing parent management group is the root management group, the permission requirements don't apply. Because the root management group is the default landing spot for all new management groups and subscriptions, you don't need permissions on it to move an item.

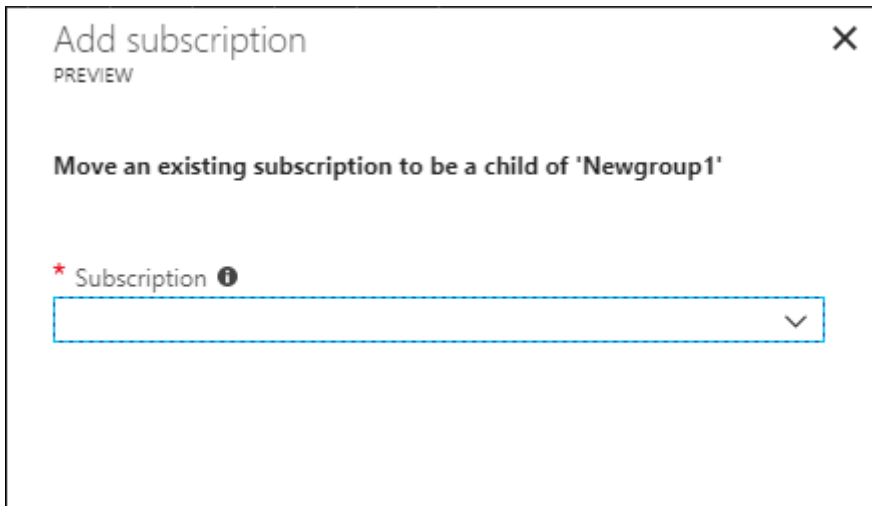
If the Owner role on the subscription is inherited from the current management group, your move targets are limited. You can move the subscription only to another management group where you have the Owner role. You can't move the subscription to

a management group where you're only a Contributor because you would lose ownership of the subscription. If you're directly assigned to the Owner role for the subscription, you can move it to any management group where you have the Contributor role.

To see what permissions you have in the Azure portal, select the management group and then select **IAM**. To learn more about Azure roles, see [What is Azure role-based access control \(Azure RBAC\)?](#).

Add an existing subscription to a management group in the portal

1. Sign in to the [Azure portal](#) [↗](#).
2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.
3. Select the management group that you want to be the parent.
4. At the top of the page, select **Add subscription**.
5. From **Add subscription** select the subscription in the list with the correct ID.



6. Select **Save**.

Remove a subscription from a management group in the portal

1. Sign in to the [Azure portal](#) [↗](#).
2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.

3. Select the management group that's the current parent.
4. Select the ellipsis (...) at the end of the row for the subscription in the list that you want to move.

TYPE	MY ROLE	
Management Group		...
Management Group		...
Management Group	Owner	...
Management Group	Owner	...

5. Select **Move**.
6. On **Move** pane, select the value for **New parent management group ID**.

Move

ID

Contoso

Resource type

ManagementGroup

* New parent management group ID

7. Select **Save**.

Move a subscription in Azure PowerShell

To move a subscription in PowerShell, you use the `New-AzManagementGroupSubscription` command:

Azure PowerShell

```
New-AzManagementGroupSubscription -GroupId 'Contoso' -SubscriptionId  
'12345678-1234-1234-1234-123456789012'
```

To remove the link between the subscription and the management group, use the `Remove-AzManagementGroupSubscription` command:

Azure PowerShell

```
Remove-AzManagementGroupSubscription -GroupId 'Contoso' -SubscriptionId  
'12345678-1234-1234-1234-123456789012'
```

Move a subscription in the Azure CLI

To move a subscription in the Azure CLI, you use the `add` command:

Azure CLI

```
az account management-group subscription add --name 'Contoso' --subscription  
'12345678-1234-1234-1234-123456789012'
```

To remove the subscription from the management group, use the `subscription remove` command:

Azure CLI

```
az account management-group subscription remove --name 'Contoso' --  
subscription '12345678-1234-1234-1234-123456789012'
```

Move a subscription in an ARM template

To move a subscription in an Azure Resource Manager template (ARM template), use the following template and deploy it at the [tenant level](#):

JSON

```
{  
  "$schema": "https://schema.management.azure.com/schemas/2019-08-  
01/managementGroupDeploymentTemplate.json#",  
  "contentVersion": "1.0.0.0",  
  "parameters": {  
    "targetMgId": {  
      "type": "string",  
      "metadata": {  
        "description": "Provide the ID of the management group that  
you want to move the subscription to."  
      }  
    },  
    "subscriptionId": {
```

```

        "type": "string",
        "metadata": {
            "description": "Provide the ID of the existing subscription
to move."
        }
    },
    "resources": [
        {
            "scope": "/",
            "type": "Microsoft.Management/managementGroups/subscriptions",
            "apiVersion": "2020-05-01",
            "name": "[concat(parameters('targetMgId'), '/',
parameters('subscriptionId'))]",
            "properties": {
            }
        }
    ],
    "outputs": {}
}

```

Or, use the following Bicep file:

Bicep

```

targetScope = 'managementGroup'

@description('Provide the ID of the management group that you want to move
the subscription to.')
param targetMgId string

@description('Provide the ID of the existing subscription to move.')
param subscriptionId string

resource subToMG 'Microsoft.Management/managementGroups/subscriptions@2020-
05-01' = {
    scope: tenant()
    name: '${targetMgId}/${subscriptionId}'
}

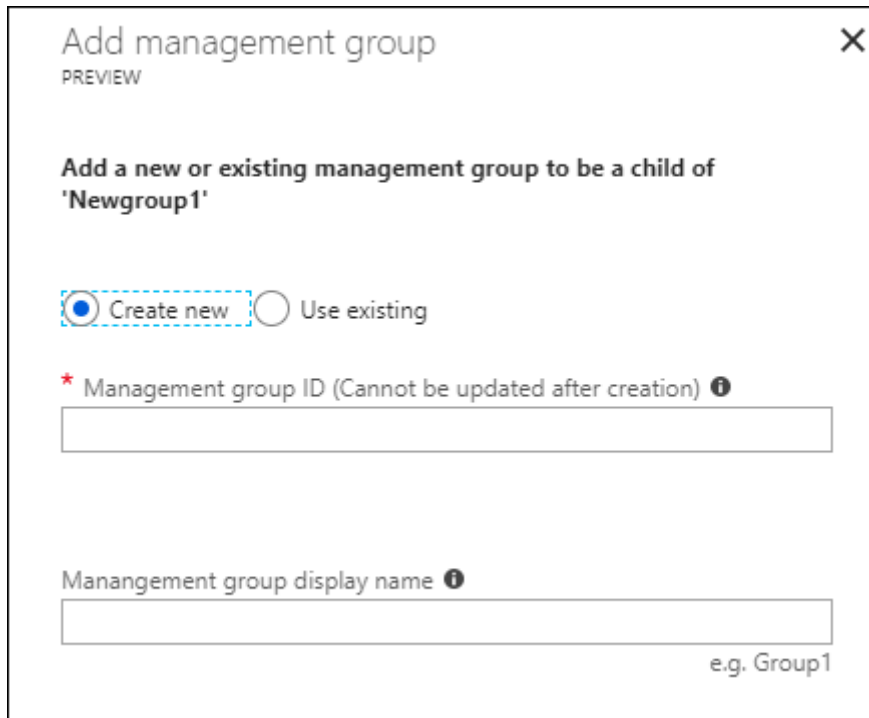
```

Move a management group in the portal

1. Sign in to the [Azure portal](#).
2. Select **All services**. In the **Filter services** text box, enter *Management groups* and select it from the list.
3. Select the management group that you want to be the parent.
4. At the top of the page, select **Create**.

5. On the **Create management group** pane, choose whether you want to use a new or existing management group:

- Selecting **Create new** creates a new management group.
- Selecting **Use existing** presents you with a dropdown list of all the management groups that you can move to this management group.



6. Select **Save**.

Move a management group in Azure PowerShell

To move a management group under a different group, use the `Update-AzManagementGroup` command in Azure PowerShell:

Azure PowerShell

```
$parentGroup = Get-AzManagementGroup -GroupId ContosoIT
Update-AzManagementGroup -GroupId 'Contoso' -ParentId $parentGroup.id
```

Move a management group in the Azure CLI

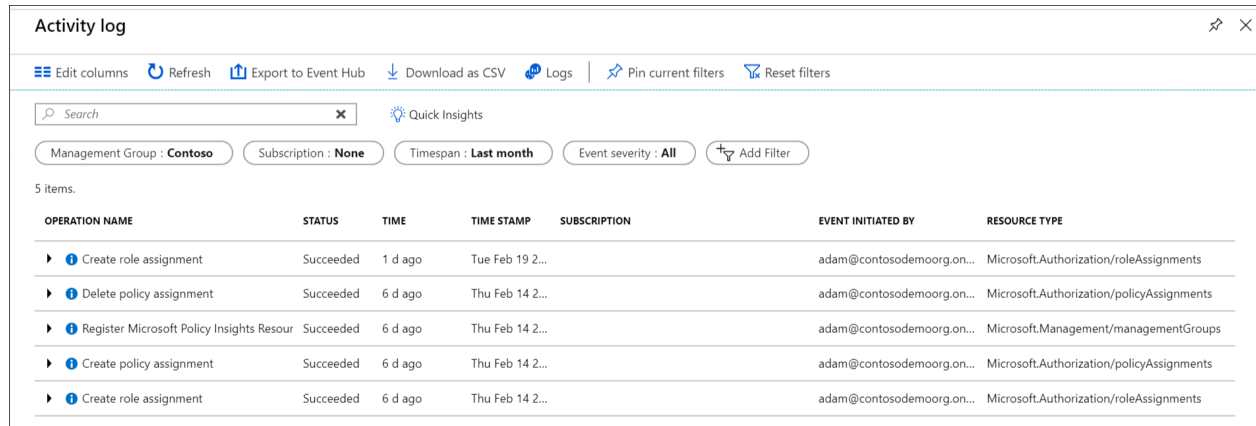
To move a management group in the Azure CLI, use the `update` command:

Azure CLI

```
az account management-group update --name 'Contoso' --parent ContosoIT
```


Audit management groups by using activity logs

Management groups are supported in [Azure Monitor activity logs](#). You can query all events that happen to a management group in the same central location as other Azure resources. For example, you can see all role assignments or policy assignment changes made to a particular management group.



The screenshot shows the 'Activity log' interface in Azure Monitor. At the top, there are navigation links: 'Edit columns', 'Refresh', 'Export to Event Hub', 'Download as CSV', 'Logs', 'Pin current filters', and 'Reset filters'. Below these is a search bar and a 'Quick Insights' link. The filters section shows 'Management Group : Contoso', 'Subscription : None', 'Timespan : Last month', and 'Event severity : All'. Below the filters, it says '5 items.' and displays a table of activity log entries.

OPERATION NAME	STATUS	TIME	TIME STAMP	SUBSCRIPTION	EVENT INITIATED BY	RESOURCE TYPE
▶ Create role assignment	Succeeded	1 d ago	Tue Feb 19 2...		adam@contosodemoorg.on...	Microsoft.Authorization/roleAssignments
▶ Delete policy assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/policyAssignments
▶ Register Microsoft Policy Insights Resour	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Management/managementGroups
▶ Create policy assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/policyAssignments
▶ Create role assignment	Succeeded	6 d ago	Thu Feb 14 2...		adam@contosodemoorg.on...	Microsoft.Authorization/roleAssignments

When you want to query on management groups outside the Azure portal, the target scope for management groups looks like

```
"/providers/Microsoft.Management/managementGroups/{yourMgID}"
```

Reference management groups from other resource providers

When you're referencing management groups from another resource provider's actions, use the following path as the scope. This path applies when you're using Azure PowerShell, the Azure CLI, and REST APIs.

```
/providers/Microsoft.Management/managementGroups/{yourMgID}
```

An example of using this path is when you're assigning a new role to a management group in Azure PowerShell:

Azure PowerShell

```
New-AzRoleAssignment -Scope  
"/providers/Microsoft.Management/managementGroups/Contoso"
```

You use the same scope path to retrieve a policy definition for a management group:

HTTP

GET

<https://management.azure.com/providers/Microsoft.Management/managementgroups/MyManagementGroup/providers/Microsoft.Authorization/policyDefinitions/ResourceNaming?api-version=2019-09-01>

Related content

To learn more about management groups, see:

- [Quickstart: Create a management group](#)
- [Review management groups in the Azure PowerShell Az.Resources module](#)
- [Review management groups in the REST API](#)
- [Review management groups in the Azure CLI](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Protect your resource hierarchy

Article • 09/06/2024

Your resources, resource groups, subscriptions, management groups, and tenant compose your resource hierarchy. Settings at the root management group, such as Azure custom roles or policy assignments, can affect every resource in your resource hierarchy. It's important to protect the resource hierarchy from changes that could negatively affect all resources.

Management groups have hierarchy settings that enable the tenant administrator to control these behaviors. This article covers each of the available hierarchy settings and how to set them.

Azure RBAC permissions for hierarchy settings

Configuring hierarchy settings requires the following resource provider operations on the root management group:

- `Microsoft.Management/managementgroups/settings/write`
- `Microsoft.Management/managementgroups/settings/read`

These operations represent Azure role-based access control (Azure RBAC) permissions. They only allow a user to read and update the hierarchy settings. They don't provide any other access to the management group hierarchy or to resources in the hierarchy.

Both of these operations are available in the Azure built-in role Hierarchy Settings Administrator.

Setting: Define the default management group

By default, a new subscription that you add in a tenant becomes a member of the root management group. If you assign policy assignments, Azure RBAC, and other governance constructs to the root management group, they immediately affect these new subscriptions. For this reason, many organizations don't apply these constructs at the root management group, even though that's the desired place to assign them. In other cases, an organization wants a more restrictive set of controls for new subscriptions but doesn't want to assign them to all subscriptions. This setting supports both use cases.

By allowing the default management group for new subscriptions to be defined, you can apply organization-wide governance constructs at the root management group. You can

define a separate management group with policy assignments or Azure role assignments that are more suited to a new subscription.

Define the default management group in the portal

1. Sign in to the [Azure portal](#) .
2. Use the search bar to search for and select **Management groups**.
3. Select the root management group.
4. Select **Settings** on the left side of the page.
5. Select the **Change default management group** button.

If the **Change default management group** button is unavailable, the cause is one of these conditions:

- The management group that you're viewing isn't the root management group.
 - Your security principal doesn't have the necessary permissions to alter the hierarchy settings.
6. Select a management group from your hierarchy, and then choose the **Select** button.

Define the default management group by using the REST API

To define the default management group by using the REST API, you must call the [Hierarchy Settings](#) endpoint. Use the following REST API URI and body format. Replace `{rootMgID}` with the ID of your root management group. Replace `{defaultGroupID}` with the ID of the management group that will become the default management group.

- REST API URI:

HTTP

PUT

<https://management.azure.com/providers/Microsoft.Management/managementGroups/{rootMgID}/settings/default?api-version=2020-05-01>

- Request body:

JSON

```
{
  "properties": {
    "defaultManagementGroup":
      "/providers/Microsoft.Management/managementGroups/{defaultGroupID}"
  }
}
```

To set the default management group back to the root management group, use the same endpoint and set `defaultManagementGroup` to a value of `/providers/Microsoft.Management/managementGroups/{rootMgID}`.

Setting: Require authorization

Any user, by default, can create new management groups in a tenant. Admins of a tenant might want to provide these permissions only to specific users, to maintain consistency and conformity in the management group hierarchy. To create child management groups, a user requires the `Microsoft.Management/managementGroups/write` operation on the root management group.

Require authorization in the portal

1. Sign in to the [Azure portal](#).
2. Use the search bar to search for and select **Management groups**.
3. Select the root management group.
4. Select **Settings** on the left side of the page.
5. Turn on the **Permissions for creating new management groups** toggle.

If the **Require write permissions for creating new management groups** toggle is unavailable, the cause is one of these conditions:

- The management group that you're viewing isn't the root management group.
- Your security principal doesn't have the necessary permissions to alter the hierarchy settings.

Require authorization by using the REST API

To require authorization by using the REST API, call the [Hierarchy Settings](#) endpoint. Use the following REST API URI and body format. This value is a Boolean, so provide either `true` or `false` for the value. A value of `true` enables this method of protecting your management group hierarchy.

- REST API URI:

HTTP

PUT

```
https://management.azure.com/providers/Microsoft.Management/managementGroups/{rootMgID}/settings/default?api-version=2020-05-01
```

- Request body:

JSON

```
{
  "properties": {
    "requireAuthorizationForGroupCreation": true
  }
}
```

To turn off the setting, use the same endpoint and set `requireAuthorizationForGroupCreation` to a value of `false`.

Azure PowerShell sample

Azure PowerShell doesn't have an `Az` command to define the default management group or to require authorization. As a workaround, you can use the REST API with the following Azure PowerShell sample:

PowerShell

```
$root_management_group_id = "Enter the ID of root management group"
$default_management_group_id = "Enter the ID of default management group (or use the same ID of the root management group)"

$body = '{
  "properties": {
    "defaultManagementGroup":
"/providers/Microsoft.Management/managementGroups/' +
$default_management_group_id + '",
    "requireAuthorizationForGroupCreation": true
  }
}'
```

```
$token = (Get-AzAccessToken).Token
$headers = @{"Authorization"= "Bearer $token"; "Content-Type"=
"application/json"}
$uri =
"https://management.azure.com/providers/Microsoft.Management/managementGroups/$root_management_group_id/settings/default?api-version=2020-05-01"

Invoke-RestMethod -Method PUT -Uri $uri -Headers $headers -Body $body
```

Related content

To learn more about management groups, see:

- [Create management groups to organize Azure resources](#)
- [Change, delete, or manage your management groups](#)

Feedback

Was this page helpful?

 Yes

 No

[Provide product feedback](#)  | [Get help at Microsoft Q&A](#)

Troubleshoot errors using management groups

Article • 09/07/2021

When you create and work with management groups you might run into errors. This article describes various general errors that might occur, and it suggests ways to resolve them.

Finding error details

Most errors are the result of an issue while running a command with management groups. When a command fails, the SDK provides details about the failure. This information indicates the issue so that it can be fixed and a later command succeeds.

General errors

Scenario: Response size too large

Issue

Customers with a large [resource hierarchy](#) may get the following message when querying the [Management Groups - Get](#) REST API with a combination of `$expand` and `$recurse` parameters:

Output

```
The response of the message was too large. Use another API or other workarounds. See https://aka.ms/mg/responsesize for more info.
```

Cause

The **Get management group** REST API doesn't return results if the payload is larger than 15 MB. This REST API is intended to get details for a single management group.

Resolution

There are several methods of dealing with a response that is too large:

- Use the [Management Groups - Get Descendants](#) REST API. This API supports pagination.
- If looking for a single management group, remove the `$expand` and `$recurse` parameters from the request to reduce the response size.

Next steps

If you didn't see your problem or are unable to solve your issue, visit one of the following channels for more support:

- Get answers from Azure experts through [Azure Forums](#).
- Connect with [@AzureSupport](#) - the official Microsoft Azure account for improving customer experience by connecting the Azure community to the right resources: answers, support, and experts.
- If you need more help, you can file an Azure support incident. Go to the [Azure support site](#) and select **Get Support**.

az account management-group

Manage Azure Management Groups.

Commands

 Expand table

Name	Description	Type	Status
az account management-group check-name-availability	Check if a Management Group Name is Valid.	Core	GA
az account management-group create	Create a new management group.	Core	GA
az account management-group delete	Delete an existing management group.	Core	GA
az account management-group entities	Entity operations (Management Group and Subscriptions) for Management Groups.	Core	GA
az account management-group entities list	List all entities for the authenticated user.	Core	GA
az account management-group hierarchy-settings	Provide operations for hierarchy settings defined at the management group level. Settings can only be set on the root Management Group of the hierarchy.	Core	GA
az account management-group hierarchy-settings create	Create hierarchy settings defined at the Management Group level.	Core	GA
az account management-group hierarchy-settings delete	Delete the hierarchy settings defined at the Management Group level.	Core	GA
az account management-group hierarchy-settings list	Get all the hierarchy settings defined at the Management Group level.	Core	GA
az account management-group hierarchy-settings update	Update the hierarchy settings defined at the Management Group level.	Core	GA
az account management-group list	List all management groups in the current tenant.	Core	GA

Name	Description	Type	Status
az account management-group show	Get the details of the management group.	Core	GA
az account management-group subscription	Subscription operations for Management Groups.	Core	GA
az account management-group subscription add	Add a subscription to a management group.	Core	GA
az account management-group subscription remove	Remove an existing subscription from a management group.	Core	GA
az account management-group subscription show	Show the details of a subscription under a known management group.	Core	GA
az account management-group subscription show-sub-under-mg	Get the subscription under a management group.	Core	GA
az account management-group tenant-backfill	Backfill Tenant Subscription Operations for Management Groups.	Core	GA
az account management-group tenant-backfill get	Get the backfill status for a tenant.	Core	GA
az account management-group tenant-backfill start	Start backfilling subscriptions for a tenant.	Core	GA
az account management-group update	Update an existing management group.	Core	GA

az account management-group check-name-availability

 Edit

Check if a Management Group Name is Valid.

Azure CLI

```
az account management-group check-name-availability --name
                                                    [--acquire-policy-token]
                                                    [--change-reference]
```

Examples

Check if a Management Group Name is Valid.

Azure CLI

```
az account management-group check-name-availability --name GroupName
```

Required Parameters

--name -n

Name of the management group.

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--acquire-policy-token

Acquiring an Azure Policy token automatically for this resource operation.

 Expand table

Property	Value
Parameter group:	Global Policy Arguments

--change-reference

The related change reference ID for this resource operation.

 Expand table

Property	Value
Parameter group:	Global Policy Arguments

Global Parameters ^

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/>  for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

az account management-group create

[Edit](#)

Create a new management group.

Azure CLI

```
az account management-group create --name  
                                   [--acquire-policy-token]  
                                   [--change-reference]  
                                   [--display-name]  
                                   [--no-register]  
                                   [--parent]
```

Examples

Create a new management group.

Azure CLI

```
az account management-group create --name GroupName
```

Create a new management group with a specific display name.

Azure CLI

```
az account management-group create --name GroupName --display-name DisplayName
```

Create a new management group with a specific parent.

Azure CLI

```
az account management-group create --name GroupName --parent ParentId/ParentName
```

Create a new management group with a specific display name and parent.

Azure CLI

```
az account management-group create --name GroupName --display-name DisplayName --parent ParentId/ParentName
```

Required Parameters

--name -n

Name of the management group.

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--acquire-policy-token

Acquiring an Azure Policy token automatically for this resource operation.

 Expand table

Property	Value
Parameter group:	Global Policy Arguments

--change-reference

The related change reference ID for this resource operation.

 Expand table

Property	Value
Parameter group:	Global Policy Arguments

--display-name -d

Sets the display name of the management group. If null, the group name is set as the display name.

--no-register

Skip registration for resource provider Microsoft.Management.

 Expand table

Property	Value
Default value:	False

--parent -p

Sets the parent of the management group. Can be the fully qualified id or the name of the management group. If null, the root tenant group is set as the parent.

Global Parameters

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/> for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

az account management-group delete

 Edit

Delete an existing management group.

Azure CLI

```
az account management-group delete --name  
                                   [--acquire-policy-token]  
                                   [--change-reference]  
                                   [--no-register]
```

Examples

Delete an existing management group

Azure CLI

```
az account management-group delete --name GroupName
```

Required Parameters

--name -n

Name of the management group.

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--acquire-policy-token

Acquiring an Azure Policy token automatically for this resource operation.

[Expand table](#)

Property	Value
Parameter group:	Global Policy Arguments

--change-reference

The related change reference ID for this resource operation.

[Expand table](#)

Property	Value
Parameter group:	Global Policy Arguments

--no-register

Skip registration for resource provider Microsoft.Management.

[Expand table](#)

Property	Value
Default value:	False

[Global Parameters](#) ^

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/>  for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

az account management-group list

 Edit

List all management groups in the current tenant.

Azure CLI

```
az account management-group list [--no-register]
```

Examples

List all management groups

Azure CLI

```
az account management-group list
```

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--no-register

Skip registration for resource provider Microsoft.Management.

 Expand table

Property	Value
Default value:	False

Global Parameters ^

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/>  for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

az account management-group show

[Edit](#)

Get the details of the management group.

Azure CLI

```
az account management-group show --name  
                                [--expand]  
                                [--no-register]  
                                [--recurse]
```

Examples

Get a management group.

Azure CLI

```
az account management-group show --name GroupName
```

Get a management group with children in the first level of hierarchy.

Azure CLI

```
az account management-group show --name GroupName -e
```

Get a management group with children in all levels of hierarchy.

Azure CLI

```
az account management-group show --name GroupName -e -r
```

Required Parameters

```
--name -n
```

Name of the management group (the last segment of the resource ID). Do not use display name.

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--expand -e

If given, lists the children in the first level of hierarchy.

 Expand table

Property	Value
Default value:	False

--no-register

Skip registration for resource provider Microsoft.Management.

 Expand table

Property	Value
Default value:	False

--recurse -r

If given, lists the children in all levels of hierarchy.

 Expand table

Property	Value
Default value:	False

Global Parameters

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/>  for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

az account management-group update

[Edit](#)

Update an existing management group.

Azure CLI

```
az account management-group update --name
                                   [--acquire-policy-token]
                                   [--add]
                                   [--change-reference]
                                   [--display-name]
                                   [--force-string]
                                   [--parent]
                                   [--remove]
                                   [--set]
```

Examples

Update an existing management group with a specific display name.

Azure CLI

```
az account management-group update --name GroupName --display-name DisplayName
```

Update an existing management group with a specific parent.

Azure CLI

```
az account management-group update --name GroupName --parent ParentId/ParentName
```

Update an existing management group with a specific display name and parent.

Azure CLI

```
az account management-group update --name GroupName --display-name DisplayName --
parent ParentId/ParentName
```

Required Parameters

--name -n

Name of the management group.

Optional Parameters

The following parameters are optional, but depending on the context, one or more might become required for the command to execute successfully.

--acquire-policy-token

Acquiring an Azure Policy token automatically for this resource operation.

[Expand table](#)

Property	Value
Parameter group:	Global Policy Arguments

--add

Add an object to a list of objects by specifying a path and key value pairs. Example: `--add property.listProperty <key=value, string or JSON string>`.

[Expand table](#)

Property	Value
Parameter group:	Generic Update Arguments
Default value:	[]

--change-reference

The related change reference ID for this resource operation.

[Expand table](#)

Property	Value
Parameter group:	Global Policy Arguments

--display-name -d

Updates the display name of the management group. If null, no change is made.

--force-string

When using 'set' or 'add', preserve string literals instead of attempting to convert to JSON.

[Expand table](#)

Property	Value
Parameter group:	Generic Update Arguments
Default value:	False

--parent -p

Update the parent of the management group. Can be the fully qualified id or the name of the management group. If null, no change is made.

--remove

Remove a property or an element from a list. Example: `--remove property.list`
`<indexToRemove>` OR `--remove propertyToRemove`.

[Expand table](#)

Property	Value
Parameter group:	Generic Update Arguments
Default value:	[]

--set

Update an object by specifying a property path and value to set. Example: `--set`
`property1.property2=<value>`.

[Expand table](#)

Property	Value
Parameter group:	Generic Update Arguments

Property	Value
Default value:	[]

Global Parameters ^

--debug

Increase logging verbosity to show all debug logs.

 Expand table

Property	Value
Default value:	False

--help -h

Show this help message and exit.

--only-show-errors

Only show errors, suppressing warnings.

 Expand table

Property	Value
Default value:	False

--output -o

Output format.

 Expand table

Property	Value
Default value:	json
Accepted values:	json, jsonc, none, table, tsv, yaml, yamlc

--query

JMESPath query string. See <http://jmespath.org/> for more information and examples.

--verbose

Increase logging verbosity. Use --debug for full debug logs.

 Expand table

Property	Value
Default value:	False

Az.Resources Module

This topic displays help topics for the Azure Resource Manager Cmdlets.

Active Directory

 [Expand table](#)

Cmdlet	Description
Add-AzADAppPermission	Adds an API permission.
Add-AzADGroupMember	Adds member to group.
Get-AzADAppCredential	Lists key credentials and password credentials for an application.
Get-AzADAppFederatedCredential	Get federatedIdentityCredentials by Id from applications.
Get-AzADApplication	Lists entities from applications or get entity from applications by key
Get-AzADAppPermission	Lists API permissions the application has requested.
Get-AzADGroup	Lists entities from groups or get entity from groups by key
Get-AzADGroupMember	Lists members from group.
Get-AzADGroupOwner	The owners of the group. Limited to 100 owners. Nullable. If this property is not specified when creating the group, the calling user is automatically assigned as the group owner. Supports \$filter (/ \$count eq 0, / \$count ne 1). Supports \$expand including nested \$select. For example, /groups? \$filter=startsWith(displayName,'Role')&\$select=id,displayName&\$expand=owners(\$select=id,user
Get-AzADOrganization	Retrieve a list of organization objects.
Get-AzADServicePrincipal	Lists entities from service principals or get entity from service principals by key
Get-AzADServicePrincipalAppRoleAssignment	Get appRoleAssignments from servicePrincipals
Get-AzADSpCredential	Lists key credentials and password credentials for an service principal.
Get-AzADUser	Lists entities from users or get entity from users by key
New-AzADAppCredential	Creates key credentials or password credentials for an application.
New-AzADAppFederatedCredential	Create federatedIdentityCredential for applications.
New-AzADApplication	Adds new entity to applications
New-AzADGroup	Adds new entity to groups
New-AzADGroupOwner	Create new navigation property ref to owners for groups
New-AzADServicePrincipal	Adds new entity to servicePrincipals
New-AzADServicePrincipalAppRoleAssignment	Create new navigation property to appRoleAssignments for servicePrincipals
New-AzADSpCredential	Creates key credentials or password credentials for an service principal.
New-AzADUser	Adds new entity to users
Remove-AzADAppCredential	Removes key credentials or password credentials for an application.
Remove-AzADAppFederatedCredential	Delete navigation property federatedIdentityCredentials for applications
Remove-AzADApplication	Deletes entity from applications
Remove-AzADAppPermission	Removes an API permission.
Remove-AzADGroup	Deletes entity from groups.
Remove-AzADGroupMember	Deletes member from group Users, contacts, and groups that are members of this group. HTTP Methods: GET (all groups), POST (supported for security groups and mail-enabled security groups), DELETE (supported for security groups) Read-only. Nullable. Supports \$expand.

Cmdlet	Description
Remove-AzADGroupOwner	Delete ref of navigation property owners for groups
Remove-AzADServicePrincipal	Deletes entity from service principal.
Remove-AzADServicePrincipalAppRoleAssignment	Delete navigation property appRoleAssignments for servicePrincipals
Remove-AzADSpCredential	Removes key credentials or password credentials for an service principal.
Remove-AzADUser	Deletes entity from users.
Update-AzADAppFederatedCredential	Update the navigation property federatedIdentityCredentials in applications
Update-AzADApplication	Updates entity in applications
Update-AzADGroup	Update entity in groups
Update-AzADServicePrincipal	Updates entity in service principal
Update-AzADServicePrincipalAppRoleAssignment	Update the navigation property appRoleAssignments in servicePrincipals
Update-AzADUser	Updates entity in users

Managed Applications

 Expand table

Cmdlet	Description
Get-AzManagedApplication	Gets managed applications
Get-AzManagedApplicationDefinition	Gets managed application definitions
New-AzManagedApplication	Creates an Azure managed application.
New-AzManagedApplicationDefinition	Creates a managed application definition.
Remove-AzManagedApplication	Removes a managed application
Remove-AzManagedApplicationDefinition	Removes a managed application definition
Set-AzManagedApplication	Updates managed application
Set-AzManagedApplicationDefinition	Updates managed application definition

Policy

 Expand table

Cmdlet	Description
Get-AzPolicyAlias	Get-AzPolicyAlias retrieves and outputs Azure provider resource types that have aliases defined and match the given parameter values. If no parameters are provided, all provider resource types that contain an alias will be output. The -ListAvailable switch modifies this behavior by listing all matching resource types including those without aliases.
Get-AzPolicyAssignment	Gets policy assignments.
Get-AzPolicyDefinition	Gets policy set definitions.
Get-AzPolicyExemption	Gets policy exemptions.
Get-AzPolicySetDefinition	Gets policy set definitions.
Get-AzRoleManagementPolicy	Get the specified role management policy for a resource scope
Get-AzRoleManagementPolicyAssignment	Get the specified role management policy assignment for a resource scope
New-AzPolicyAssignment	Creates or updates a policy assignment.

Cmdlet	Description
New-AzPolicyDefinition	Creates or updates a policy definition.
New-AzPolicyExemption	Creates or updates a policy exemption.
New-AzPolicySetDefinition	Creates or updates a policy set definition.
New-AzRoleManagementPolicyAssignment	Create a role management policy assignment
Remove-AzPolicyAssignment	This operation deletes a policy assignment, given its name and the scope it was created in. The scope of a policy assignment is the part of its ID preceding '/providers/Microsoft.Authorization/policyAssignments/{policyAssignmentName}'.
Remove-AzPolicyDefinition	This operation deletes the policy definition in the given subscription with the given name.
Remove-AzPolicyExemption	This operation deletes a policy exemption, given its name and the scope it was created in. The scope of a policy exemption is the part of its ID preceding '/providers/Microsoft.Authorization/policyExemptions/{policyExemptionName}'.
Remove-AzPolicySetDefinition	This operation deletes the policy set definition in the given subscription with the given name.
Remove-AzRoleManagementPolicy	Delete a role management policy
Remove-AzRoleManagementPolicyAssignment	Delete a role management policy assignment
Update-AzPolicyAssignment	This operation updates a policy assignment with the given scope and name. Policy assignments apply to all resources contained within their scope. For example, when you assign a policy at resource group scope, that policy applies to all resources in the group.
Update-AzPolicyDefinition	This operation updates an existing policy definition in the given subscription or management group with the given name.
Update-AzPolicyExemption	This operation updates a policy exemption with the given scope and name.
Update-AzPolicySetDefinition	This operation updates an existing policy set definition in the given subscription or management group with the given name.
Update-AzRoleManagementPolicy	Update a role management policy
Update-AzRoleManagementPolicyAssignment	Update a role management policy assignment

Resources

 [Expand table](#)

Cmdlet	Description
Export-AzResourceGroup	Captures a resource group as a template and saves it to a file.
Export-AzTemplateSpec	Exports a Template Spec to the local filesystem
Get-AzDataBoundaryScope	Get data boundary at specified scope
Get-AzDataBoundaryTenant	Get data boundary of tenant.
Get-AzDenyAssignment	Lists Azure RBAC deny assignments at the specified scope. By default it lists all deny assignments in the selected Azure subscription. Use respective parameters to list deny assignments to a specific user, or to list deny assignments on a specific resource group or resource. The cmdlet may call below Microsoft Graph API according to input parameters: • GET /directoryObjects/{id} • POST /directoryObjects/getByIds
Get-AzDeployment	Get deployment

Cmdlet	Description
Get-AzDeploymentOperation	Get deployment operation
Get-AzDeploymentScript	Gets or lists deployment scripts.
Get-AzDeploymentScriptLog	Gets the log of a deployment script execution.
Get-AzDeploymentWhatIfResult	Gets a template What-If result for a deployment at subscription scope.
Get-AzLocation	Gets all locations and the supported resource providers for each location.
Get-AzManagementGroup	Gets Management Group(s)
Get-AzManagementGroupDeployment	Get deployment at a management group
Get-AzManagementGroupDeploymentOperation	Get deployment operation for management group deployment
Get-AzManagementGroupDeploymentStack	Gets Management Group scoped Deployment Stacks.
Get-AzManagementGroupDeploymentWhatIfResult	Gets a template What-If result for a deployment at management group scope.
Get-AzManagementGroupEntity	Lists all Entities under the current Tenant
Get-AzManagementGroupHierarchySetting	Gets the Hierarchy Settings under the current tenant
Get-AzManagementGroupNameAvailability	Checks if the Management Group name is available in the Tenant and a valid name.
Get-AzManagementGroupSubscription	Gets the details of Subscription(s) under a Management Group.
Get-AzPrivateLinkAssociation	Gets all the Azure Resource Management Private Link Association(s).
Get-AzProviderFeature	Gets information about Azure provider features.
Get-AzProviderOperation	Gets the operations for an Azure resource provider that are securable using Azure RBAC.
Get-AzProviderPreviewFeature	Gets a feature registration in your account.
Get-AzResource	Gets resources.
Get-AzResourceGroup	Gets resource groups.
Get-AzResourceGroupDeployment	Gets the deployments in a resource group.
Get-AzResourceGroupDeploymentOperation	Gets the resource group deployment operation
Get-AzResourceGroupDeploymentStack	Gets Resource Group scoped Deployment Stacks.
Get-AzResourceGroupDeploymentWhatIfResult	Gets a template What-If result for a deployment at resource group scope.
Get-AzResourceLock	Gets a resource lock.
Get-AzResourceManagementPrivateLink	Gets Azure Resource Management Private Link(s)
Get-AzResourceProvider	Gets a resource provider.
Get-AzRoleAssignment	Lists Azure RBAC role assignments at the specified scope. By default it lists all role assignments in the selected Azure subscription. Use respective parameters to list assignments to a specific user, or to list assignments on a specific resource group or resource. The cmdlet may call below Microsoft Graph API according to input parameters: • GET /users/{id} • GET /servicePrincipals/{id} • GET /groups/{id} • GET /directoryObjects/{id} • POST /directoryObjects/getByIds

Cmdlet	Description
	Please notice that this cmdlet will mark <code>ObjectType</code> as <code>Unknown</code> in output if the object of role assignment is not found or current account has insufficient privileges to get object type.
Get-AzRoleAssignmentSchedule	Get the specified role assignment schedule for a resource scope
Get-AzRoleAssignmentScheduleInstance	Gets the specified role assignment schedule instance.
Get-AzRoleAssignmentScheduleRequest	Get the specified role assignment schedule request.
Get-AzRoleDefinition	Lists all Azure RBAC roles that are available for assignment.
Get-AzRoleEligibilitySchedule	Get the specified role eligibility schedule for a resource scope
Get-AzRoleEligibilityScheduleInstance	Gets the specified role eligibility schedule instance.
Get-AzRoleEligibilityScheduleRequest	Get the specified role eligibility schedule request.
Get-AzRoleEligibleChildResource	Get the child resources of a resource on which user has eligible access
Get-AzSubscriptionDeploymentStack	Gets Subscription scoped Deployment Stacks.
Get-AzTag	Gets predefined Azure tags Gets the entire set of tags on a resource or subscription.
Get-AzTemplateSpec	Gets or lists Template Specs
Get-AzTenantBackfillStatus	Get the current Tenant Backfill Subscription Status
Get-AzTenantDeployment	Get deployment at tenant scope
Get-AzTenantDeploymentOperation	Get deployment operation for deployment at tenant scope
Get-AzTenantDeploymentWhatIfResult	Gets a template What-If result for a deployment at tenant scope.
Invoke-AzResourceAction	Invokes an action on a resource.
Move-AzResource	Moves a resource to a different resource group or subscription.
New-AzDenyAssignment	Creates a user-assigned deny assignment at the specified scope. By default, the deny assignment targets Everyone and requires at least one excluded principal. Alternatively, use <code>-PrincipalId</code> and <code>-PrincipalType</code> to target a specific user or service principal.
New-AzDeployment	Create a deployment at the current subscription scope.
New-AzManagementGroup	Creates a Management Group
New-AzManagementGroupDeployment	Create a deployment at a management group
New-AzManagementGroupDeploymentStack	Creates a new Management Group scoped Deployment Stack.
New-AzManagementGroupHierarchySetting	Creates Hierarchy Settings under the current tenant
New-AzManagementGroupSubscription	Adds a Subscription to a Management Group.
New-AzPrivateLinkAssociation	Creates the Azure Resource Management Private Link Association.
New-AzResource	Creates a resource.
New-AzResourceGroup	Creates an Azure resource group.
New-AzResourceGroupDeployment	Adds an Azure deployment to a resource group.
New-AzResourceGroupDeploymentStack	Creates a new Resource Group scoped Deployment Stack.

Cmdlet	Description
New-AzResourceLock	Creates a resource lock.
New-AzResourceManagementPrivateLink	Create Azure Resource Management Private Link
New-AzRoleAssignment	Assigns the specified RBAC role to the specified principal, at the specified scope. The cmdlet may call below Microsoft Graph API according to input parameters: • GET /users/{id} • GET /servicePrincipals/{id} • GET /groups/{id} • GET /directoryObjects/{id} Please notice that this cmdlet will mark <code>ObjectType</code> as <code>Unknown</code> in output if the object of role assignment is not found or current account has insufficient privileges to get object type.
New-AzRoleAssignmentScheduleRequest	Create a role assignment schedule request.
New-AzRoleDefinition	Creates a custom role in Azure RBAC. Provide either a JSON role definition file or a PSRoleDefinition object as input. First, use the Get-AzRoleDefinition command to generate a baseline role definition object. Then, modify its properties as required. Finally, use this command to create a custom role using role definition.
New-AzRoleEligibilityScheduleRequest	Create a role eligibility schedule request.
New-AzSubscriptionDeploymentStack	Creates a new Subscription scoped Deployment Stack.
New-AzTag	Creates a predefined Azure tag or adds values to an existing tag Creates or updates the entire set of tags on a resource or subscription.
New-AzTemplateSpec	Creates a new Template Spec.
New-AzTenantDeployment	Create a deployment at tenant scope
Publish-AzBicepModule	Publishes a Bicep file to a registry.
Register-AzProviderFeature	Registers an Azure provider feature in your current subscription context.
Register-AzProviderPreviewFeature	Creates a feature registration in your account.
Register-AzResourceProvider	Registers a resource provider.
Remove-AzDenyAssignment	Removes a user-assigned deny assignment at the specified scope.
Remove-AzDeployment	Removes a deployment and any associated operations
Remove-AzDeploymentScript	Removes a deployment script and its associated resources.
Remove-AzManagementGroup	Removes a Management Group
Remove-AzManagementGroupDeployment	Removes a deployment at a management group and any associated operations
Remove-AzManagementGroupDeploymentStack	Removes a Management Group scoped Deployment Stack.
Remove-AzManagementGroupHierarchySetting	Deletes all Hierarchy Settings under the current tenant
Remove-AzManagementGroupSubscription	Removes a Subscription from a Management Group.
Remove-AzPrivateLinkAssociation	Delete a specific azure private link association.
Remove-AzResource	Removes a resource.
Remove-AzResourceGroup	Removes a resource group.
Remove-AzResourceGroupDeployment	Removes a resource group deployment and any

Cmdlet	Description
	associated operations.
Remove-AzResourceGroupDeploymentStack	Removes a Resource Group scoped Deployment Stack.
Remove-AzResourceLock	Removes a resource lock.
Remove-AzResourceManagementPrivateLink	Deletes the Resource Management Private Link.
Remove-AzRoleAssignment	Removes a role assignment to the specified principal who is assigned to a particular role at a particular scope. The cmdlet may call below Microsoft Graph API according to input parameters: • GET /users/{id} • GET /servicePrincipals/{id} • GET /groups/{id} • GET /directoryObjects/{id} • POST /directoryObjects/getByIds Please notice that this cmdlet will mark <code>ObjectType</code> as <code>Unknown</code> in output if the object of role assignment is not found or current account has insufficient privileges to get object type.
Remove-AzRoleDefinition	Deletes a custom role in Azure RBAC. The role to be deleted is specified using the Id property of the role. Delete will fail if there are existing role assignments made to the custom role.
Remove-AzSubscriptionDeploymentStack	Removes a Subscription scoped Deployment Stack.
Remove-AzTag	Deletes predefined Azure tags or values Deletes the entire set of tags on a resource or subscription.
Remove-AzTemplateSpec	Removes a Template Spec
Remove-AzTenantDeployment	Removes a deployment at tenant scope and any associated operations
Save-AzDeploymentScriptLog	Saves the log of a deployment script execution to disk.
Save-AzDeploymentTemplate	Saves a deployment template to a file.
Save-AzManagementGroupDeploymentStackTemplate	Saves a Management Group scoped Deployment Stack Template.
Save-AzManagementGroupDeploymentTemplate	Saves a deployment template to a file.
Save-AzResourceGroupDeploymentStackTemplate	Saves a Resource Group scoped Deployment Stack Template.
Save-AzResourceGroupDeploymentTemplate	Saves a resource group deployment template to a file.
Save-AzSubscriptionDeploymentStackTemplate	Saves a Subscription scoped Deployment Stack Template.
Save-AzTenantDeploymentTemplate	Saves a deployment template to a file.
Set-AzDataBoundary	Opt-in tenant to data boundary.
Set-AzManagementGroupDeploymentStack	Sets a new Management Group scoped Deployment Stack.
Set-AzResource	Modifies a resource.
Set-AzResourceGroup	Modifies a resource group.
Set-AzResourceGroupDeploymentStack	Sets a new Resource Group scoped Deployment Stack.
Set-AzResourceLock	Modifies a resource lock.
Set-AzRoleAssignment	Update an existing Role Assignment. The cmdlet may call below Microsoft Graph API according to input parameters: • GET /users/{id}

Cmdlet	Description
	• GET /servicePrincipals/{id} • GET /groups/{id} • GET /directoryObjects/{id} • POST /directoryObjects/getByIds Please notice that this cmdlet will mark <code>ObjectType</code> as <code>Unknown</code> in output if the object of role assignment is not found or current account has insufficient privileges to get object type.
Set-AzRoleDefinition	Modifies a custom role in Azure RBAC. Provide the modified role definition either as a JSON file or as a PSRoleDefinition. First, use the Get-AzRoleDefinition command to retrieve the custom role that you wish to modify. Then, modify the properties that you wish to change. Finally, save the role definition using this command.
Set-AzSubscriptionDeploymentStack	Sets a new Subscription scoped Deployment Stack.
Set-AzTemplateSpec	Modifies a Template Spec.
Start-AzTenantBackfill	Starts backfilling subscriptions for the current Tenant
Stop-AzDeployment	Cancel a running deployment
Stop-AzManagementGroupDeployment	Cancel a running deployment at a management group
Stop-AzResourceGroupDeployment	Cancels a resource group deployment.
Stop-AzRoleAssignmentScheduleRequest	Cancels a pending role assignment schedule request.
Stop-AzRoleEligibilityScheduleRequest	Cancels a pending role eligibility schedule request.
Stop-AzTenantDeployment	Cancel a running deployment at tenant scope
Test-AzDeployment	Validates a deployment.
Test-AzManagementGroupDeployment	Validates a deployment at a management group.
Test-AzManagementGroupDeploymentStack	Validates a management group scoped deployment stack.
Test-AzResourceGroupDeployment	Validates a resource group deployment.
Test-AzResourceGroupDeploymentStack	Validates a resource group scoped deployment stack.
Test-AzSubscriptionDeploymentStack	Validates a subscription scoped deployment stack.
Test-AzTenantDeployment	Validates a deployment at tenant scope.
Unregister-AzProviderFeature	Unregisters an Azure provider feature in your account.
Unregister-AzProviderPreviewFeature	Removes a feature registration from your account.
Unregister-AzResourceProvider	Unregisters a resource provider.
Update-AzManagementGroup	Updates a Management Group
Update-AzManagementGroupHierarchySetting	Updates Hierarchy Settings under the current tenant
Update-AzRoleAssignmentScheduleRequest	Update a role assignment schedule request.
Update-AzRoleEligibilityScheduleRequest	Update a role eligibility schedule request.
Update-AzTag	Selectively updates the set of tags on a resource or subscription.

ManagementGroups interface

Package: [@azure/arm-managementgroups](#)

Interface representing a ManagementGroups.

Methods

[Expand table](#)

beginCreateOrUpdate(string, CreateManagementGroupRequest, ManagementGroupsCreateOrUpdateOptionalParams)	Create or update a management group. If a management group is already created and a subsequent create request is issued with different properties, the management group properties will be updated.
beginCreateOrUpdateAndWait(string, CreateManagementGroupRequest, ManagementGroupsCreateOrUpdateOptionalParams)	Create or update a management group. If a management group is already created and a subsequent create request is issued with different properties, the management group properties will be updated.
beginDelete(string, ManagementGroupsDeleteOptionalParams)	Delete management group. If a management group contains child resources, the request will fail.
beginDeleteAndWait(string, ManagementGroupsDeleteOptionalParams)	Delete management group. If a management group contains child resources, the request will fail.
get(string, ManagementGroupsGetOptionalParams)	Get the details of the management group.
list(ManagementGroupsListOptionalParams)	List management groups for the authenticated user.
listDescendants(string, ManagementGroupsGetDescendantsOptionalParams)	List all entities that descend from a management group.
update(string, PatchManagementGroupRequest, ManagementGroupsUpdateOptionalParams)	Update a management group.

Method Details

beginCreateOrUpdate(string, CreateManagementGroupRequest, ManagementGroupsCreateOrUpdateOptionalParams)

Create or update a management group. If a management group is already created and a subsequent create request is issued with different properties, the management group

properties will be updated.

TypeScript

```
function beginCreateOrUpdate(groupId: string, createManagementGroupRequest:
CreateManagementGroupRequest, options?:
ManagementGroupsCreateOrUpdateOptionalParams):
Promise<PollerLike<PollOperationState<ManagementGroup>, ManagementGroup>>
```

Parameters

groupId string

Management Group ID.

createManagementGroupRequest [CreateManagementGroupRequest](#)

Management group creation parameters.

options [ManagementGroupsCreateOrUpdateOptionalParams](#)

The options parameters.

Returns

Promise<[PollerLike](#)<[@azure/core-lro.PollOperationState](#)<[ManagementGroup](#)>, [ManagementGroup](#)>>

beginCreateOrUpdateAndWait(string, CreateManagementGroupRequest, ManagementGroupsCreateOrUpdateOptionalParams)

Create or update a management group. If a management group is already created and a subsequent create request is issued with different properties, the management group properties will be updated.

TypeScript

```
function beginCreateOrUpdateAndWait(groupId: string, createManagementGroupRequest:
CreateManagementGroupRequest, options?:
ManagementGroupsCreateOrUpdateOptionalParams): Promise<ManagementGroup>
```

Parameters

groupId string

Management Group ID.

createManagementGroupRequest [CreateManagementGroupRequest](#)

Management group creation parameters.

options [ManagementGroupsCreateOrUpdateOptionalParams](#)

The options parameters.

Returns

Promise<[ManagementGroup](#)>

beginDelete(string, ManagementGroupsDeleteOptionalParams)

Delete management group. If a management group contains child resources, the request will fail.

TypeScript

```
function beginDelete(groupId: string, options?:  
ManagementGroupsDeleteOptionalParams):  
Promise<PollerLike<PollOperationState<ManagementGroupsDeleteResponse>,  
ManagementGroupsDeleteResponse>>
```

Parameters

groupId string

Management Group ID.

options [ManagementGroupsDeleteOptionalParams](#)

The options parameters.

Returns

Promise<[PollerLike](#)<@azure/core-Iro.PollOperationState<[ManagementGroupsDeleteResponse](#)>,
[ManagementGroupsDeleteResponse](#)>>

beginDeleteAndWait(string, ManagementGroupsDeleteOptionalParams)

Delete management group. If a management group contains child resources, the request will fail.

TypeScript

```
function beginDeleteAndWait(groupId: string, options?: ManagementGroupsDeleteOptionalParams): Promise<ManagementGroupsDeleteResponse>
```

Parameters

groupId string

Management Group ID.

options [ManagementGroupsDeleteOptionalParams](#)

The options parameters.

Returns

Promise<[ManagementGroupsDeleteResponse](#)>

get(string, ManagementGroupsGetOptionalParams)

Get the details of the management group.

TypeScript

```
function get(groupId: string, options?: ManagementGroupsGetOptionalParams): Promise<ManagementGroup>
```

Parameters

groupId string

Management Group ID.

options [ManagementGroupsGetOptionalParams](#)

The options parameters.

Returns

Promise<[ManagementGroup](#)>

list(ManagementGroupsListOptionalParams)

List management groups for the authenticated user.

TypeScript

```
function list(options?: ManagementGroupsListOptionalParams):  
  PagedAsyncIterableIterator<ManagementGroupInfo, ManagementGroupInfo[],  
  PageSettings>
```

Parameters

options [ManagementGroupsListOptionalParams](#)

The options parameters.

Returns

[PagedAsyncIterableIterator](#)<[ManagementGroupInfo](#), [ManagementGroupInfo](#)[], [PageSettings](#)>

listDescendants(string, ManagementGroupsGetDescendantsOptionalParams)

List all entities that descend from a management group.

TypeScript

```
function listDescendants(groupId: string, options?:  
  ManagementGroupsGetDescendantsOptionalParams):  
  PagedAsyncIterableIterator<DescendantInfo, DescendantInfo[], PageSettings>
```

Parameters

groupId string

Management Group ID.

options [ManagementGroupsGetDescendantsOptionalParams](#)

The options parameters.

Returns

update(string, PatchManagementGroupRequest, ManagementGroupsUpdateOptionalParams)

Update a management group.

TypeScript

```
function update(groupId: string, patchGroupRequest: PatchManagementGroupRequest, options?: ManagementGroupsUpdateOptionalParams): Promise<ManagementGroup>
```

Parameters

groupId string

Management Group ID.

patchGroupRequest [PatchManagementGroupRequest](#)

Management group patch parameters.

options [ManagementGroupsUpdateOptionalParams](#)

The options parameters.

Returns

Promise<[ManagementGroup](#)>

managementgroups Package

Packages

 Expand table

aio
models
operations

Classes

 Expand table

ManagementGroupsAPI	The Azure Management Groups API enables consolidation of multiple subscriptions/resources into an organizational hierarchy and centrally manage access control, policies, alerting and reporting for those resources.
-------------------------------------	---

Management Groups

Article • 10/31/2023

Management groups enable you to manage access, policies, and compliance for your Azure subscriptions. For an introduction, see [What are Azure management groups?](#).

See also

- [What are Azure management groups?](#)
- [Quickstart: Create a management group with REST API](#)

Microsoft.Management managementGroups

Remarks

When deployed at tenant scope, don't set the scope property. See create management groups with tenant deployments in [Bicep file](#) or [ARM templates](#).

When deployed at other scopes, set the scope property to `tenant()` for Bicep files or `/` for ARM templates. See create management groups with management group deployments in [Bicep file](#) or [ARM templates](#).

Bicep resource definition

The managementGroups resource type can be deployed with operations that target:

- **Tenant** - See [tenant deployment commands](#)

For a list of changed properties in each API version, see [change log](#).

Resource format

To create a Microsoft.Management/managementGroups resource, add the following Bicep to your template.

Bicep

```
resource symbolicname 'Microsoft.Management/managementGroups@2024-02-01-preview' = {
  name: 'string'
  properties: {
    details: {
      parent: {
        id: 'string'
      }
    }
    displayName: 'string'
  }
}
```

Property Values

Microsoft.Management/managementGroups

 Expand table

Name	Description	Value
name	The resource name	string (required)
properties	The generic properties of a management group used during creation.	CreateManagementGroupPropertiesOrManagementGroupProperties

CreateManagementGroupDetailsOrManagementGroupDetails

 Expand table

Name	Description	Value
parent	(Optional) The ID of the parent management group used during creation.	CreateParentGroupInfoOrParentGroupInfo

CreateManagementGroupPropertiesOrManagementGroupPropertie

 Expand table

Name	Description	Value
details	The details of a management group used during creation.	CreateManagementGroupDetailsOrManagementGroupDetails
displayName	The friendly name of the management group. If no value is passed then this field will be set to the groupId.	string

CreateParentGroupInfoOrParentGroupInfo

 Expand table

Name	Description	Value
id	The fully qualified ID for the parent management group. For example, /providers/Microsoft.Management/managementGroups/00000000-0000-0000-0000-000000000000	string

Usage Examples

Azure Verified Modules

The following [Azure Verified Modules](#) can be used to deploy this resource type.

 Expand table

Module	Description
Management Group	AVM Resource Module for Management Group

Last updated on 01/20/2026